NONUNIQUE ERGODICITY ON THE BOUNDARY OF OUTER SPACE

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Abstract. To an $\mathbb{R}$-tree in the boundary of Outer space, we associate two simplices: the simplex of projective length measures, and the simplex of projective dual currents. For both kinds of simplices, we estimate the dimension of maximal simplices for arational $\mathbb{R}$-trees in the boundary of Outer space.

Introduction

Given a complete hyperbolic surface S of finite area and a fixed geodesic lamination Λ on S, one can consider the space of measures transverse to Λ. This space of measures naturally has the structure of a simplicial cone, and so, when considered projectively, sits as a simplex inside the boundary of the Teichmüller space of the surface, $\mathcal{PML}(S)$. The vertices of this simplex are exactly the projective classes of ergodic measures which are transverse to Λ. The most interesting case to consider is when Λ is a *filling* geodesic lamination. In [12], Gabai shows that given k nonisotopic pairwise disjoint simple closed curves on S, one can construct a filling geodesic lamination on S which admits k distinct projective classes of ergodic measures. On the other hand, Levitt [22] shows that the number of pants curves on S gives the upper bound of the number of ergodic measures supported on noncompact leaves of a foliation on S using a cohomological argument. Moreover, Lenzhen and Masur [21] show that any filling geodesic lamination on S which admits k projective classes of ergodic measures can be approximated by a sequence of multicurves with k simple closed curve components. It then follows that the maximal number of distinct projective classes of ergodic measures transverse to a filling geodesic lamination on S is precisely the number of curves in a pants decomposition of S.

In this paper, we investigate the corresponding question for trees in the boundary of Culler–Vogtmann's Outer space, CV_n [11]. The boundary of CV_n consists of the set of *very small F_n-trees* which are either not free or not discrete. For such F_n-trees, the most interesting case to consider is when the tree is *arational*, meaning the restriction of the F_n action to any proper free factor is free and discrete. On the other hand, the F_n-trees come equipped with a natural length measure. Projective classes of length measures on an F_n-tree form a simplex in ∂CV_n, with vertices of this simplex corresponding to ergodic ones. When the simplex is a point, we say T is **uniquely ergometric**. The first examples of nonuniquely ergometric arational trees, not dual to a measured lamination on a surface, were constructed by Martin [24]. The first systematic study on the nonunique ergometricity of F_n-trees was done by Guirardel [14, Corollary 5.4]. We estimate the maximum nonunique ergometricity for arational trees.

Theorem A. *For $n \geq 5$, the maximal number of projective classes of ergodic length functions on an arational F_n-tree is in the interval $[2n-6, 2n-1]$.*

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One can also consider the space of projective *currents* ν that are dual to an F_n-tree T, in the sense that the length pairing $\langle T, \nu \rangle = 0$. This space again forms a simplex, with the vertices of this simplex representing ergodic currents. If the simplex of currents dual to T is a point, then we say the tree T is **uniquely ergodic**. Nonunique ergodicity of F_n-trees was first studied by Coulbois–Hilion–Lustig [8, Section 7]. We give a similar estimate for the maximum nonunique ergodicity for arational trees.

Theorem B. *For $n \geq 5$, the maximal number of projective classes of ergodic currents dual to an arational F_n-tree is in the interval $[2n-6, 2n-1]$.*

For smaller cases of n, we make the following remarks. When $n = 1$, Outer space CV_1 is a point, so the boundary ∂CV_1 is empty. For $n = 2$, any arational $\mathbb{R}$-tree is geometric, in the sense that it arises as the dual tree of a filling measured lamination on a punctured torus. Since the set of minimal filling laminations is homeomorphic to $\mathbb{R} \setminus \mathbb{Q}$ for a punctured torus (see e.g. [13]), any measured filling lamination is uniquely ergodic. Hence, when $n = 2$, any arational tree is uniquely ergometric and uniquely ergodic. When $n = 3$ or 4, we can obtain a better lower bound of $n - 1$ by using a rose graph with positive train track structure following an idea similar to one presented in Sections 2 and 3.

Given a measured geodesic lamination on a surface, there is an $\mathbb{R}$-tree which is dual to this lamination. When the fundamental group of the surface is free of rank n, this tree represents a point in ∂CV_n. Trees in ∂CV_n which arise as the dual tree to a measured geodesic lamination on a surface are said to be *of surface type*. The space of ergodic currents dual to a surface-type tree coincides precisely with the space of ergodic length measures on the tree, and the bounds given by Gabai [12], Lenzhen–Masur [21], and Levitt [22] hold. By Reynolds [27], any *geometric* arational F_n-tree T is dual to a filling measured geodesic lamination on a surface S with one boundary component. Say $S = S_{g,1}$ has genus g and one boundary component, so that T is an F_n-tree with $n = 2g$. As a pants decomposition of S contains $3g - 2$ curves, we see that the maximum number of projectively distinct ergodic length measures (or ergodic currents) is $3g-2$, which is on the order of $\frac{3}{2}n$. Thus, our results show that nongeometric arational trees can admit more ergodic length measures and ergodic currents, on the order of $2n$.

It is still an open question whether or not the simplex of projective classes of measured currents dual to an F_n-tree T always has the same dimension as the simplex of projective classes of length functions on T for arational trees which are nongeometric.

Our approach is inspired by the work of Namazi–Pettet–Reynolds [26]. For the lower bound in Theorem A, we consider a folding path in CV_n so that the transition matrices satisfy conditions where diagonal entries dominate others. We phrase these conditions conveniently in terms of an "Alice and Bob game," see Theorem 2.3. The difficulty here is in finding a folding path that realizes these conditions, and we suspect that our bound of $2n-6$ can be improved. For Theorem B, the lower bound is obtained similarly, but instead using unfolding paths in CV_n. In both constructions, a technical difficulty lies in ensuring that the limiting trees are arational. We do this by enumerating free factors and guiding the (un)folding path to ensure that these factors act discretely in the limit, one at a time, while ensuring that the matrix conditions still hold.

We remark that the matrices we construct have deep connections to the field of dynamics and have been used to build examples of non-uniquely ergodic systems.

For instance, in [20], Keane constructed a minimal 4-interval exchange with two ergodic measures, and Yoccoz [29, Section 8.3] constructed minimal interval exchange transformations with the maximal number of ergodic measures.

For the upper bound in both Theorem A and Theorem B, we use Transverse Decomposition theorems, Theorems 1.16 and 1.17, to obtain a nice transverse decomposition of graphs induced from linearly independent ergodic measures. Namazi–Pettet–Reynolds [26] gave the first construction for such a transverse decomposition, but we give a more direct argument. We then put an order on the edges with respect to the decomposition so that adding the edges in the increasing order strictly raises the "complexity" of the resultant graph. Since the number of steps to obtain the whole graph depends only on the number of different ergodic components from the transverse decomposition, this gives an upper bound on the number of ergodic measures.

In Section 1, we give the necessary background on Outer space, length measures, currents, folding/unfolding sequences, arationality and ergodicity. We also give our proofs of the Transverse Decomposition theorems, Theorems 1.16 and 1.17. Then, toward the proof of Theorem A, Section 2 establishes the lower bound $2n-6$ for the maximal number of projective classes of length measures on an $\mathbb{R}$-tree, T. This bound is obtained by constructing T as a limiting tree of a folding sequence whose transition matrices have diagonal-dominant entries. In Section 3, we begin our proof of Theorem B by establishing the same lower bound for the maximal number of projective classes of measured currents dual to an $\mathbb{R}$-tree in the accumulation set of an unfolding sequence. We do this by constructing an unfolding sequence with diagonal-dominant transition matrices, which is a bit more involved than the folding sequence construction. In Section 4, we provide the other bound for Theorem A and Theorem B by establishing the upper bound $2n-1$ for both simplices of nonuniquely ergodic and nonuniquely ergometric trees using the ergodic transverse decomposition and the complexity of graphs. To finalize the proofs of Theorem A and Theorem B, we prove in Section 5 that we can arrange the folding and unfolding sequences constructed in Section 2 and Section 3 to ensure that the limiting objects are arational and non-geometric, while preserving the number of linearly independent measures.

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Contents

1. Preliminaries

1.1. **Graphs and Outer space.** A **graph** is a CW complex of dimension at most 1. Denote by R_n the rose graph with n petals, which is a graph with one 0-cell (called a **vertex**) and n 1-cells (called **edges**). For a graph G, we denote by VG the set of vertices of G and EG the set of edges of G. A **marking** of a graph G of rank n is a homotopy equivalence $m : R_n \to G$. A **metric** on G is a function $\ell : EG \to \mathbb{R}_{>0}$, which assigns to each edge a positive number. Indeed, ℓ endows a path metric on G: the distance between two points is defined as the length of the shortest (weighted) path joining the two vertices. The **volume**, $\mathrm{vol}(G, \ell)$, of a metric graph (G, ℓ) is the sum $\sum_{e \in EG} \ell(e)$. Culler–Vogtmann [11] introduced **Outer space**, CV_n, as the space of equivalence classes of finite marked metric graphs of rank n and of unit volume:

$$CV_n := \{(G, m, \ell) \mid \mathrm{vol}(G, \ell) = 1\}/\sim,$$

where $(G, m, \ell) \sim (G', m', \ell')$ if there is an isometry $i : G \to G'$ such that $i \circ m$ is homotopic to m'. The collection of marked metric graphs of rank n without the unit volume condition is denoted by cv_n, the **unprojectivized Outer space**. Culler–Vogtmann also showed CV_n is contractible. For a more detailed exposition on CV_n, see [1].

Let G be a finite graph of rank n such that all vertices have valence at least 3, and fix a marking $m : R_n \to G$. Given any other metric graph G', a map $f : G \to G'$

is said to be a **morphism** if it maps vertices to vertices and it is locally injective on the interior of every edge (and hence has no backtrackings). The **transition matrix** of a morphism $f : G \to G'$ is the matrix which records how the morphism f maps edges of G over edges of G'. In particular, if we label the (unoriented) edges of G by $e_1, e_2, \dots, e_m$ and the edges of G' by $e'_1, e'_2, \dots, e'_n$, then the transition matrix M_f for the morphism $f : G \to G'$ is the $n \times m$ matrix where the ij-entry records the number of times the path $f(e_j)$ crosses over the edge e'_i. Per Stallings [28], we have the following definitions.

Definition 1.1. Given a graph G and two edges e, e' of G, a **fold** is the natural quotient morphism $G \to G/(e \sim e')$. Moreover, if e, e' only share one vertex, then the fold is said to be of **type I**. Otherwise, if e, e' share both vertices, then the fold is said to be of **type II**.

Both types of folds induce surjective homomorphisms on fundamental groups, but only type I folds induce injective ones. Stallings [28, Algorithm 5.4] proved that any morphism which is a homotopy equivalence can be decomposed as a sequence of type I folds.

1.2. **Train tracks and the geometry of Outer space.** For a Lipschitz map f between two metric spaces, we denote by $\sigma(f)$ its Lipschitz constant. We can now define the distance between two marked metric graphs $(G, m, \ell), (G', m', \ell')$ in CV_n as follows.

$$d(G, G') := \inf \left\{ \log \sigma(f) \mid f : G \to G' \text{ is Lipschitz and } fm \simeq m' \right\}.$$

This metric is called the **Lipschitz metric** on CV_n, but it is asymmetric: $d(G, G')$ may not be equal to $d(G', G)$. By the Arzéla–Ascoli theorem, the infimum above is realized for every $G, G' \in CV_n$. We will call such a witness $f : G \to G'$ an **optimal map** if, in addition to realizing the infimum, it is linear on each edge. For an optimal map $f : G \to G'$ with $\sigma(f) = \lambda$, we define the **tension graph** of f, denoted by Δ_f, to be the subgraph of G induced by those edges $e \in EG$ with slope λ (Namely, the union of $e \in EG$ such that $\sigma(f|_e) = \lambda$.)

Now we define a train track structure on a graph. For each vertex v in a graph G, a **direction** at v is an oriented edge of G with initial vertex v. Denote by $T_v(G)$ the set of all directions at v, which can be regarded as the unit tangent space of G at v. Then we give an equivalence relation on $T_v(G)$. A **train track structure** on a graph G is a collection $\{(T_v(G), \sim_v)\}_{v \in VG}$, where $\sim_v$ is an equivalence relation on $T_v(G)$ for each vertex v of G. The equivalence classes are called **gates**. A **turn** at v is an unordered pair of directions at v, and it is called **legal** if the directions belong to different gates. An immersed path in G is called **legal** if it makes legal turns at each vertex. Given a morphism $f : G \to G'$ and a vertex v in G, we note that f induces a map $Df : T_v(G) \to T_{f(v)}(G')$, as f maps vertices to vertices. We then have a natural train track structure on G induced by f where for each vertex v of G, we define an equivalence relation $\sim_v$ on $T_v(G)$ as:

$$d_1 \sim_v d_2 \quad \text{if and only if} \quad Df(d_1) = Df(d_2).$$

Given a morphism $f : G \to G'$, we will always consider the induced train track structure on G, unless otherwise specified. With this induced structure, we notice that an immersed edge path p in G is legal if $f(p)$ is a reduced path, i.e., f is locally injective on the vertices of p (as well as on the interior of the edges).

1.3. **Folding and unfolding sequences.** A **folding/unfolding** sequence is a sequence of graphs,

$$\cdots \to G_{-2} \to G_{-1} \to G_0 \to G_1 \to G_2 \to \cdots,$$

together with morphisms $\{\psi_s : G_s \to G_{s+1}\}$ that are homotopy equivalences, and so that all finite compositions are morphisms. We consider two special cases: **folding** sequences, which are enumerated by nonnegative integers and denoted by $(G_s)_{s\geq 0}$, and **unfolding** sequences, enumerated by nonpositive integers and denoted by $(G_s)_{s\leq 0}$.

For each $r, s \in \mathbb{Z}$ with $r < s$, we write the composition of maps as

$$\phi_{r,s} := \psi_{s-1} \circ \cdots \circ \psi_{r+1} \circ \psi_r : G_r \to G_s$$

For each morphism $\phi_{r,s} : G_r \to G_s$, denote by $M = M^{r,s}$ its transition matrix. Note that as we compose functions right to left, we have that $\phi_{r,t} = \phi_{s,t} \circ \phi_{r,s}$ and $M^{r,t} = M^{s,t} \cdot M^{r,s}$. We say that an unfolding sequence $(G_s)_{s\leq 0}$ is **reduced** if there is no proper invariant subgraph in the sequence. That is, if there is a sequence $\{H_s\}_{s\leq 0}$ of subgraphs $H_t \subseteq G_t$ for $t \leq 0$ and there is $s_0 < 0$ such that $\phi_{r,s}(H_r) \subseteq H_s$ for $r < s \leq s_0$, then either $H_s = G_s$ for all $-s$ or H_s contains no edges for all large $-s$. Likewise, a folding sequence $(G_s)_{s\geq 0}$ is **reduced** if for any sequence $\{H_s\}_{s\geq 0}$ of subgraphs $H_t \subseteq G_t$ for $t \geq 0$, if there is $r_0 > 0$ such that $\phi_{r,s}(H_r) \subseteq H_s$ for all $s > r \geq r_0$, then either $H_r = G_r$ for all large r or H_r contains no edges for all r.

In a folding sequence, train track structures on any G_r induced by the morphisms $G_r \to G_s$ eventually stabilize as $s \to \infty$, and in practice we construct folding sequences by fixing a train track structure on G_0 and then folding compatibly.

A **length measure** on a folding/unfolding sequence $(G_s)_s$ is defined to be a sequence of vectors $(\vec{\ell}_s)_s$ in $\mathbb{R}^{|EG_s|}$, where for each s, the vectors satisfy the compatibility condition ('lengths pull back'):

$$\vec{\ell}_s = (M^{s,s+1})^T \vec{\ell}_{s+1}.$$

We consider the set $\mathcal{L}((G_s)_{s\geq 0})$ of length measures on the folding sequence $(G_s)_{s\geq 0}$ as the cone of non-negative vectors in the finite dimensional vector space whose vectors are sequences $(\vec{v}_s)_{s\geq 0}$ with $\vec{v}_s \in \mathbb{R}^{|EG_s|}$ for every $s \geq 0$ satisfying the compatibility condition

$$\vec{v}_s = (M^{s,s+1})^T \vec{v}_{s+1}$$

for every $s \geq 0$. The dimension of this space is nonzero and bounded above by $\liminf_{s\to\infty} |EG_s| > 0$.

Recall that a morphism that is a homotopy equivalence can be realized as a composition of type I folds, after possibly subdividing, (see Section 1.1). Then the lengths of conjugacy classes in F_n are nonincreasing in the folding sequence. Hence, seeing $\overline{CV_n} \subset \mathbb{PR}^{F_n}_{\geq 0}$ (see Section 1.4), it follows that a folding sequence $(G_s)_{s\geq 0}$ always projectively converges to an $\mathbb{R}$-tree T, which we call the **limit tree**. More precisely, to a reduced folding sequence $(G_s)_{s\geq 0}$, Namazi–Pettet–Reynolds associate a *topological limit tree* $\hat{T}$, [26, Section 5.1]. This is a topological tree (a space that is uniquely path connected and locally path connected) admitting an F_n-action obtained by considering the universal cover T_0 of G_0, identifying points in T_0 whose images coincide in the universal cover T_n of G_n, for some $n > 0$, and finally ensuring that this quotient space is Hausdorff. They show that the natural F_n-action on $\hat{T}$ has dense orbits. Now, assigning a length measure to the folding sequence $(G_s)_{s\geq 0}$ yields a pseudo-metric on $\hat{T}$ as follows. For each $s \geq 0$, denote by d_s the induced metric on the universal cover $T_s := \widetilde{G_s}$. Then, $\hat{T}$ has a pseudo-metric defined as

$$d(x,y) := \inf_{s\geq 0} d_s(x_s, y_s),$$

where $x_s, y_s \in T_s$ are points mapped to $x, y \in \hat{T}$, respectively, under the quotient map $T_s \to \hat{T}$. Finally, let T be the $\mathbb{R}$-tree obtained from $\hat{T}$ by identifying pairs of points at pseudo-distance 0, and let d be its induced metric. Then,

Lemma 1.2 ([26, Lemma 5.1])**.** *A folding sequence $(G_s, \vec{\ell}_s)_{s\geq 0}$ of marked metric graphs, after projection to CV_n, converges to an $\mathbb{R}$-tree (T, d), called the **limit tree**, where d is the induced metric obtained from the pseudo-metric on the topological limit tree $\hat{T}$ of the folding sequence.*

Recall a **length measure** on an F_n-$\mathbb{R}$-tree T is a collection of finite Borel measures $\{\mu_I \mid I \text{ is a compact interval of } T\}$ such that $\mu_J = (\mu_I)|_J$ for all $J \subset I$. It is F_n-equivariant if $\mu_{g(I)} = (g|_I)_*\mu_I$ for all I and $g \in F_n$. The following theorem formalizes the relationship described above between the set of length measures on the limiting tree T to those on the folding sequence $(G_s)_{s\geq 0}$.

Theorem 1.3 ([26, Proposition 5.4])**.** *Given a reduced folding sequence $(G_s)_{s\geq 0}$ with limiting tree T, there is a natural linear isomorphism between the space of length measures on $(G_s)_{s\geq 0}$ and the space of equivariant length measures on T.*

A nonzero length measure on the limit tree $\hat{T}$ may well assign 0 to a nondegenerate interval, so the associated $\mathbb{R}$-tree is obtained from $\hat{T}$ by collapsing subtrees.

Dual to any length measure is a *width measure*. A **width measure** (or a **current**) on a folding/unfolding sequence $(G_s)_s$ is a sequence of non-negative vectors $(\vec{\mu}_s)_s$ in $\mathbb{R}^{|EG_s|}$ where for each s, the vectors satisfy the compatibility condition ('currents push forward'):

$$\vec{\mu}_s = M^{s-1,s}\vec{\mu}_{s-1}.$$

We consider the set $\mathcal{C}((G_s)_{s\leq 0})$ of width measures on the unfolding sequence $(G_s)_{s\leq 0}$ as the cone of non-negative vectors in the finite-dimensional vector space whose vectors are sequences $(\vec{v}_s)_{s\leq 0}$ with $\vec{v}_s \in \mathbb{R}^{|EG_s|}_{\geq 0}$ for every $s \leq 0$ satisfying the compatibility condition

$$\vec{v}_s = M^{s-1,s}\vec{v}_{s-1}$$

for every $s \leq 0$. The dimension of this space is nonzero and bounded above by $\liminf_{s\to -\infty} |EG_s| > 0$.

Unlike the case of folding sequences, the lengths of graphs in an unfolding sequence increase. Hence, an unfolding sequence does not necessarily converge to a tree, but after rescaling, it may have a nontrivial accumulation set. Instead of this accumulation set for an unfolding sequence, we consider a *legal lamination* defined as follows. In an unfolding sequence of graphs, $(G_s)_{s\leq 0}$, we say a path p in G_s is **legal** if it is legal with respect to $\phi_{s,0} : G_s \to G_0$ (equivalently, $\phi_{s,0}$ immerses p in G_0). Denote by $\Omega^L_\infty(G_s)$ the set of bi-infinite legal paths in G_s. Now define the **legal lamination** Λ of an unfolding sequence $(G_s)_{s\leq 0}$ as:

$$\Lambda := \bigcap_{s\leq 0} \phi_{s,0}(\Omega^L_\infty(G_s)),$$

which is a subset of $\Omega^L_\infty(G_0)$. Similar to the limiting tree of a folding sequence, the set of *currents* (see Section 1.5) on the legal lamination characterizes the set of width measures on the unfolding sequence:

Theorem 1.4 ([26, Theorem 4.4])**.** *Given a reduced unfolding sequence $(G_s)_{s\leq 0}$ with legal lamination Λ, there is a natural linear isomorphism between the space of width measures supported on $(G_s)_{s\leq 0}$ and the space of currents supported on Λ.*

We sketch the proof of Theorem 1.4 at the end of Section 1.5 as it is in unpublished work.

1.4. **Boundary of Outer space and laminations.** Let ∂F_n be the Gromov boundary of the Cayley graph of F_n, which is well-defined up to homeomorphism. Then define the **double boundary** $\partial^2 F_n$ as $\partial F_n \times \partial F_n$ minus its diagonal. It is equipped with a natural **flip** action denoted by $i : \partial^2 F_n \to \partial^2 F_n$, which swaps the two factors. We further observe that for a finitely-generated subgroup $H \leq F_n$, we can embed $\partial^2 H \subset \partial^2 F_n$ as H is quasiconvex in F_n. Moreover, the action of F_n on the Cayley graph of F_n by homeomorphisms extends to ∂F_n and hence to $\partial^2 F_n$. We define a **lamination** L as a nonempty, closed, F_n-invariant, flip-invariant subset of $\partial^2 F_n$.

We now want to change our perspective on Outer space. Recall each element of CV_n is a marked metric graph (G, m, ℓ) of unit volume. Then the universal cover of G is a tree, T. Denoting the covering map by $p : T \to G$, we can define the length function $\widetilde{\ell}$ on the set of edges of T by letting each edge e of T have length $\widetilde{\ell}(e) := \ell(p(e))$. The length function $\widetilde{\ell}$ then extends to a path metric d_T on T. Note that by construction, $\widetilde{\ell}$ is equivariant under the F_n-covering space action. Because the covering space action is discrete and free, we can regard each element in CV_n as an F_n-tree with a discrete, free action by isometries. Conversely, one has to restrict the set of F_n-trees to **minimal** ones, namely those for which there is no F_n-invariant proper subtree. Then we can view CV_n as:

$$CV_n = \{F_n\text{-tree } T \mid \text{discrete, free, minimal, and isometric } F_n\text{-action}\}/\sim,$$

where $T \sim T'$ if and only if there is an F_n-equivariant homothety between two F_n-trees T and T'. Weakening the equivalence condition to F_n-equivariant isometry gives cv_n, the unprojectivized Outer space (See Section 1.1.)

This new perspective on CV_n helps us to discuss its boundary. Indeed, Culler–Morgan [10] showed that minimal F_n-trees are characterized by the associated **translation length function** $\ell_T : g \mapsto \inf\{d_t(x, g\cdot x) \mid x \in T\}$ on F_n. Therefore, with the F_n-tree perspective on CV_n, we can embed CV_n into $\mathbb{R}^{F_n}_{\geq 0}$, which stays embedded in the projectivization $\mathbb{PR}^{F_n}_{\geq 0}$. As the image of CV_n in $\mathbb{PR}^{F_n}_{\geq 0}$ has a compact closure, we can define the boundary as $\partial CV_n := \overline{CV_n} \setminus CV_n$. The points on the boundary also correspond to F_n-trees, but they are either not free or not discrete. The F_n-trees in the compactification $\overline{CV_n}$ are collectively characterized as *very small* trees, by Bestvina–Feighn [2], Cohen–Lustig [6], and Horbez [17].

Like in the boundary of Teichmüller space, we can associate a lamination $L(T)$ to a point T on the boundary of Outer space. We construct $L(T)$ as follows. From the F_n-action on the Cayley graph of F_n, each nontrivial element $g \in F_n$ has a repelling fixed point $g^{-\infty}$ and an attracting fixed point g^{∞} on the boundary ∂F_n. We define the **dual lamination** $L(T)$ of T as:

$$L(T) := \bigcap_{m=1}^{\infty} \overline{\left\{(g^{-\infty}, g^{\infty}) \in \partial^2 F_n \mid \ell_T(g) < \frac{1}{m}\right\}},$$

which is indeed a lamination by construction. Intuitively, $L(T)$ records which elements of F_n act on T with arbitrarily small translation length. Each element of

$L(T)$ is called a **leaf**. We say a finitely-generated subgroup H of F_n **carries a leaf of** $L(T)$, if there is a leaf $\ell \in L(T)$ such that $\ell \in \partial^2 H$. (Recall that $\partial^2 H \subset \partial^2 F_n$.) This is equivalent to saying that either some element of H fixes a point in T (thus having an element of translation length 0), or the restricted action of H to a minimal H-invariant subtree $T_H \subset T$ is not discrete (thus having elements whose translation lengths converge to zero).

A train track structure on a graph is called **recurrent** if there is a legal loop that crosses every edge. The following lemma will be used to prove Theorem 1.10.

Lemma 1.5 ([5, Lemma 7.1])**.** *Let $T \in CV_n$ and let Σ be a fixed simplex in CV_n. Then there exists a point T_0 in Σ such that every optimal map $f : T_0 \to T$ induces a recurrent train track structure on T_0/F_n.*

1.5. **Arational limiting objects.** An $\mathbb{R}$-tree $T \in \overline{CV_n}$ is **arational** if for every proper free factor $H \leq F_n$, the F_n action restricted to H is free and discrete. According to Reynolds [27, Theorem 1.1], there are two kinds of arational trees. The **geometric** arational trees are dual to a filling measured lamination on a punctured surface (so the conjugacy class of the puncture is elliptic). The **non-geometric** arational trees are free. A lamination L is called **arational** if no leaf of L is carried by a proper free factor of F_n. Hence, $T \in \partial CV_n$ is arational if $L(T)$ is arational. Bestvina–Reynolds [5, Theorem 1.1] proved the boundary of the free factor complex, FF_n, for F_n is homeomorphic to the quotient space of the space of arational trees by equating trees with the same dual lamination. For $n \geq 3$, the free factor complex is the simplicial complex with vertices corresponding to the conjugacy classes of nontrivial proper free factors of F_n and, for $k = 1, \ldots, n-2$, with k-simplices $[A_0, \ldots, A_k]$ corresponding to the filtration $A_0 < A_1 < \ldots < A_k$ of proper containment up to conjugation. When $n = 2$, we modify the edge relation so that A, B are joined by an edge if and only if A, B form a basis for F_2.

We consider a coarsely well-defined projection $\pi : cv_n \to FF_n$, given by $\pi(T)$ for $T \in cv_n$ as the collection of free factors represented by subgraphs of T/F_n. Note that the map π factors through CV_n.

The usefulness of this projection π to FF_n is that it sends geodesics in cv_n (so, in particular, folding paths) to quasi-geodesics in FF_n, up to reparametrization. A **folding path** (or **unfolding path**) is a collection of metric graphs G_t parametrized by $t \in [0, \infty]$ ($t \in (-\infty, 0]$ respectively), such that for $t < t'$ we have maps $f_{tt'} : G_t \to G_{t'}$ and for $t < t' < t''$, we have $f_{tt''} = f_{t't''} f_{tt'}$ such that $f_{tt'}$ are isometries on edges, but do not necessarily send vertices to vertices. Then a folding sequence (or unfolding sequence) we defined earlier in Section 1.3 is just a discrete subset of a folding path (unfolding path, respectively) such that vertices are mapped to vertices.

More precisely, for $K > 0$ and a metric space X, we say a function $f : [0, \infty) \to X$ is a **reparametrized K-quasi-geodesic** if there is a (finite or infinite) sequence $0 = t_0 < t_1 < \ldots < \infty$ such that $\mathrm{diam}(f([t_i, t_{i+1}])) \leq K$ for all i and $|i - j| \leq d(f(t_i), f(t_j)) + 2$, and either:

- the sequence (t_i) is infinite and $t_i \to \infty$, or
- the sequence (t_i) is finite and $\mathrm{diam}(f([t_m, \infty))) \leq K$ where t_m is the largest term in the sequence.

In other words, f is a K-quasi-geodesic after some reparametrization by t_i's. Then as stated above, we have

Proposition 1.6 ([3, Corollary 6.5 and Proposition 9.2])**.** *Let G_t be a folding path in cv_n. Then $\pi(G_t)$ in FF_n is a reparametrized K-quasi-geodesic with K depending only on n.*

To study arational trees further, we consider another 'measure' on an $\mathbb{R}$-tree other than length: a *current*. Define a **current** as an F_n-invariant and flip-invariant Radon measure μ on $\partial^2 F_n$. See [23, 18] for an introduction to currents on free groups. Denote by $\mathrm{Curr}(F_n)$ the set of currents, and by $\mathbb{P}\,\mathrm{Curr}(F_n)$ the set of projective classes of currents.

For each element $g \in F_n$, we can construct a current $\eta_g \in \mathrm{Curr}(F_n)$ as follows. First, suppose g is **primitive**, that is, there is no element $h \in F_n$ and $m \geq 2$ such that $g = h^m$. Denote by $[g]$ the conjugacy class of g. Then we define

$$\eta_g := \sum_{h\in[g]} \left(\delta_{(h^{-\infty},h^{\infty})} + \delta_{(h^{\infty},h^{-\infty})}\right),$$

where $\delta_{(x,y)}$ denotes the Dirac measure concentrated at the point $(x, y) \in \partial^2 F_n$: namely, for $A \subset \partial^2 F_n$:

$$\delta_{(x,y)}(A) = \begin{cases} 1 & \text{if } (x,y) \in A, \\ 0 & \text{otherwise.} \end{cases}$$

Now for a non-primitive element $g = h^m$ for some $h \in F_n$ and $m \geq 2$, we define $\eta_g := m\eta_h$. Then we call η_g the **counting current** given by g.

Kapovich–Lustig [19, Theorem A] constructed the $\mathrm{Out}(F_n)$-invariant continuous pairing, the **intersection form**,

$$\langle -, - \rangle : \overline{cv_n} \times \mathrm{Curr}(F_n) \to \mathbb{R}_{\geq 0},$$

which is $\mathbb{R}_{\geq 0}$-homogeneous in the first coordinate, and $\mathbb{R}_{\geq 0}$-linear in the second coordinate. Moreover, $\langle T, \eta_g \rangle = \ell_T(g)$ for $T \in cv_n$ and all counting currents η_g. The intersection form relates $\mathbb{R}$-trees and currents as dual objects.

Denote by M_n the unique minimal closed invariant subset of $\mathrm{Curr}(F_n)$ under the $\mathrm{Out}(F_n)$-action [24, Section 5.8]. Now, for each F_n-tree $T \in \overline{cv_n}$, we define T^* as the set of currents dual to T, namely:

$$T^* := \{\mu \in M_n \mid \langle T, \mu \rangle = 0\}.$$

Here we collect some properties of T^*.

Lemma 1.7 ([5, Theorem 4.4 and Remark 4.6])**.** *Let $T, T_1, T_2 \in \overline{cv_n}$. Then the following hold.*

- *(i) $T^* = \{0\}$ if and only if $T \notin \partial cv_n$.*
- *(ii) T is arational if and only if $T^* \neq \{0\}$ and any two nonzero $\mu_1, \mu_2 \in T^*$ have the same support.*
- *(iii) Suppose T_1 is arational and $T_1^* \subseteq T_2^*$. Then T_2 is arational, $T_1^* = T_2^*$ and $L(T_1) = L(T_2)$.*
- *(iv) Suppose T_2 is arational and $\{0\} \neq T_1^* \subseteq T_2^*$. Then T_1 is arational, $T_1^* = T_2^*$ and $L(T_1) = L(T_2)$.*

Note that for any trees $T_1, T_2 \in \overline{cv_n}$ which are projectively equivalent, $T_1^* = T_2^*$. In particular, the property of a tree being arational is scale-invariant by Lemma 1.7(ii). Hence it is reasonable to consider the simplices of arational trees lying in $\overline{CV_n}$.

It is convenient to call two arational $\mathbb{R}$-trees T_1, T_2 **equivalent** if $T_1^* = T_2^*$, or, equivalently, if $L(T_1) = L(T_2)$. An equivalence class of arational trees forms a simplex in ∂CV_n, and the subspace of arational trees with the equivalence classes identified to a point is homeomorphic to the boundary of the free factor complex, (see [5] and [15]). Furthermore, equivalent arational trees are equivariantly homeomorphic with respect to the observers' topology, (see [5, Proposition 3.2] and [15, Proposition 7.1]).

In Theorem 1.10 below, we realize arational trees as limits of folding sequences. For the proof, we begin with producing a *folding path* G_t converging to the desired arational tree. When a folding path G_t in cv_n converges to a tree $T \in \overline{CV_n}$, we mean the path given by the projection $\hat{G}_t$ of G_t to CV_n converges to T. Then, in Lemma 1.11, we will modify the folding path so that when restricted to a discrete set of times we obtain a folding sequence.

The following two lemmas are the tools from Bestvina–Reynolds [5] that we need to produce the desired folding path.

Lemma 1.8 ([5, Theorem 6.6], U-turn lemma)**.** *Let $\{[S_i, T_i]\}_{i\geq 0}$ be a sequence of folding paths in CV_n beginning at S_i and ending at T_i, and $U_i \in [S_i, T_i]$ be a point on the path. Suppose $S_i \to S \in \overline{CV_n}$, $T_i \to T \in \overline{CV_n}$ and $U_i \to U \in \overline{CV_n}$. Then either $U^* \subset T^*$, or $S^* \cap U^* \neq \{0\}$.*

Lemma 1.9 ([5, Lemma 7.3], realizing folding path)**.** *Let $S_i, T_i \in CV_n$ for $i \geq 0$, and assume that*

(a) all S_i are in the same simplex as S_0,
(b) there is a morphism $f_i : S_i \to T_i$ such that the induced train track structure on S_i is recurrent, and
(c) $T_i \to T \in \partial CV_n$.

If T is arational, then $S_i \to S \in CV_n$ and certain initial segments of the folding path γ_i induced by f_i converge uniformly on compact sets to a folding path γ from S that converges to $U \in \partial CV_n$ with $U^* \subseteq T^*$.

Now we are ready to prove that every arational tree on the boundary of Outer space is *reachable* by a folding sequence.

Theorem 1.10. *For each arational tree T on ∂CV_n, there exists a reduced folding sequence limiting to T.*

Proof. Let $T \in \partial CV_n$ be arational. Then there exists a sequence $\{T_i\}_{i\geq 0}$ in CV_n converging to T. Fix a simplex Σ in CV_n represented by a marked rose (R_n, m) with n petals. By Lemma 1.5, for each T_i we can find an F_n-tree $S_i \in \Sigma$ with an optimal map $f_i : S_i \to T_i$ inducing a recurrent train track structure on S_i/F_n. Let c_i be a *recurrent loop* in S_i/F_n; i.e. a legal loop in S_i/F_n crossing every edge of S_i/F_n. As S_i/F_n is a finite graph, so it can have at most finitely many train track structures. Hence, we may pass to a subsequence and assume $\{S_i/F_n\}_{i\geq 0}$ have the same train track structure and the same recurrent loop. Taking a further subsequence, we may assume S_i converges to $S \in \overline{CV_n}$. We will show $S \in CV_n$ by proving that none of the petals in S_i/F_n have lengths limiting to 0 or ∞. To that end, first note that having a recurrent loop implies the length of each petal in S_i/F_n does not blow up to infinity, before normalizing. On the other hand, if a petal's length limits to zero, the element of F_n represented by it would have length 0 in T, which contradicts the assumption that T is arational. All in all, $S \notin \partial CV_n$, so $S \in CV_n$. In particular, by Lemma 1.7(i), it follows that $S^* = \{0\}$.

Now, each optimal map $f_i : S_i \to T_i$ induces a folding path $[S_i, T_i]$. Then by Lemma 1.9, we can find a folding path γ from S converging to some $U \in \partial CV_n$ and with $U^* \subseteq T^*$. By Lemma 1.7(iv), we have that U is arational and equivalent to

T. Then, Lemma 1.11 promotes this folding path to a reduced folding sequence. Finally, by Theorem 1.3, modifying the length measures on the reduced folding sequence $(U_i)_{i\geq 0}$ converging to U, we obtain a reduced folding sequence converging to T. $\square$

Lemma 1.11. *Suppose G_t is a folding path converging to an arational tree T. Then there is a reduced folding sequence converging to T.*

Proof. Let $T \in \overline{CV_n}$ be arational and let G_t be a folding path converging to T. First note that for any point $v_t \in G_t$, there is some $t' > t$ such that $f_{tt'}(v_t) \in G_{t'}$ is a vertex. Indeed, otherwise, the images of v_t would define a free splitting of F_n which does not change along the folding path. In particular, the path is eventually constant in the free factor complex (and even in the free splitting complex) so the limiting tree cannot be arational ([5, Corollary 4.5].) Similarly, for every t there is $t' > t$ so that every edge of G_t maps onto $G_{t'}$, for otherwise we would have proper subgraphs $H_t \subset G_t$ that are preserved by $f_{tt'}$ and eventually contain loops, so the path would be coarsely constant in FF_n and would not converge to an arational tree.

Call a vertex of G_t **rigid** if it has at least 3 gates. Images of rigid vertices along the folding path are rigid vertices. First, suppose that G_t eventually contains at least one rigid vertex. For a given t, choose $t' \gg t$ so that $G_{t'}$ contains a rigid vertex v and, further, that the preimage of v intersects every open edge of G_t. Note that the second condition will hold for some such t' by the eventual surjectivity of the maps $f_{tt'}$ when restricted to each edge of G_t. Now, fold each illegal turn based at non-rigid vertices of G_t until either the folding reaches a point in the preimage of v, or else the illegal turn becomes legal and thus creates another rigid vertex. We call this folded graph H. This folding procedure gives a factorization of $G_t \to G_{t'}$ through $G_t \to H \to G_{t'}$, where the second map is a morphism which maps every vertex to a rigid vertex. Iterating this process for $G_{t'} \to G_{t''}$ for some $t'' > t'$, the sequence of graphs H produced in this way is the desired folding sequence.

Now, suppose no G_t has any rigid vertex. Pick a point $v_0 \in G_0$ and let $v_t \in G_t$ be its image in G_t. Again by the first paragraph there is a sequence $t_i \to \infty$ so that v_{t_i} is a vertex. Moreover, after passing to a subsequence, the preimage of $v_{t_{i+1}}$ in G_{t_i} intersects every open edge. Fold every illegal turn of G_{t_i} not based at v_{t_i} until a point in the preimage of $v_{t_{i+1}}$ is reached. This time, an illegal turn cannot become legal as that would create a rigid vertex. We again get a factorization as above where now all vertices of H map to $v_{t_{i+1}}$ and this gives the desired sequence.

In either case, the folding sequence is necessarily reduced, again by the first paragraph. $\square$

Similarly, we prove that every lamination dual to an arational tree is realizable as the *diagonal closure* of the legal lamination of an unfolding sequence. Here we first relate the legal lamination in terms of trees in the accumulation set of the unfolding sequence.

Lemma 1.12 ([26, Page 5])**.** *Let $(G_s)_{s\leq 0}$ be an unfolding sequence with the legal lamination Λ. Then for any tree T in the accumulation set of $(G_s)_{s\leq 0}$,*

$$\Lambda \subseteq L(T).$$

For a subset $X \subset \partial^2 F_n$, we say a leaf $\ell = (g, h) \in \partial^2 F_n$ is **diagonal over** X if there exist $p > 1$ and $x_1, \ldots, x_{p-1} \in \partial F_n$ such that

$$(g, x_1), (x_1, x_2), \ldots, (x_{p-1}, h) \in X.$$

We say X is **diagonally closed** if every leaf in $\partial^2 F_n$ diagonal over X is contained in X. We remark that dual laminations of trees are diagonally closed [8, Page 751], but legal laminations may not be.

A result of Coulbois–Hilion–Reynolds [9, Theorem A], shows that the dual lamination of an indecomposable F_n-tree is minimal up to diagonal leaves. That is, if there is a sublamination $\Lambda \subseteq L(T)$, then $\overline{\Lambda} = L(T)$, where $\overline{\Lambda}$ denotes the diagonal closure of Λ. As every arational tree is indecomposable [27, Theorem 1.1], we have the following lemma which states that the dual lamination of an arational tree is *almost minimal*.

Lemma 1.13. *Let T be an arational tree in ∂CV_n. Then $L(T)$ is minimal up to diagonal leaves.*

Proof. An arational tree is indecomposable by Reynolds [27, Theorem 1.1], and an indecomposable tree has dual lamination that is minimal up to diagonal leaves by [9, Theorem A]. □

We are now ready to show that the dual lamination of an arational tree can be realized as the diagonal closure of the legal lamination for an unfolding sequence.

Theorem 1.14. *Let Λ be a lamination that is dual to an arational tree T. Then there exists a reduced unfolding sequence whose legal lamination's diagonal closure is Λ.*

Proof. Let $\Lambda = L(T)$ for some arational tree T. Then $T \in \partial CV_n$, so we can take a sequence of trees $(T_i)_{i\geq 0}$ in CV_n converging to T. Now fix a tree $S \in CV_n$. Using Lemma 1.5, for each $i \geq 0$ we can find T_i' from the simplex of CV_n containing T_i such that there is an optimal map $f_i : T_i' \to S$ inducing a recurrent train track structure on T_i'. Then both $(T_i)_{i\geq 0}$ and $(T_i')_{i\geq 0}$ project to the same sequence of points in the free factor complex converging to a point on the boundary. Hence, denoting by T' the limit of $(T_i')_{i\geq 0}$, we have that T and T' lie in the same simplex in ∂CV_n, so T' is arational. Now let γ_i be the folding (also unfolding, as it is a finite path) path induced by $f_i : T_i' \to S$. We claim:

Claim (Realizing unfolding path)**.** *After taking a subsequence of $(\gamma_i)_{i\leq 0}$, it converges uniformly on compact sets to an unfolding path (ray) γ to S with accumulation set containing T'.*

The claim will conclude our proof, as the legal lamination Λ' of an induced unfolding sequence from γ has to be contained in $L(T')$ by Lemma 1.12. Since $L(T')$ is diagonally closed, the diagonal closure $\overline{\Lambda'}$ is also contained in $L(T')$. Then we can conclude $\overline{\Lambda'} = L(T') = L(T)$ by Lemma 1.13.

Hence, it suffices to prove the claim. Giving the natural parametrization on $(\gamma_i)_{i\geq 0}$, followed by the reparametrization so that the terminal points of γ_i's are parametrized by 0, we can write $\gamma_i : [t_i, 0] \to \overline{CV_n}$. We then extend each folding path γ_i by a constant, so that we can rewrite $\gamma_i : (-\infty, 0] \to \overline{CV_n}$ as an unfolding ray. Note that $(\gamma_i)_{i\geq 0}$ is pointwise bounded by the fact that the function $d_S : \tau \in CV_n \mapsto d(\tau, S)$ is proper. Indeed, for each $u \in (-\infty, 0]$, we have $K = d_S^{-1}([0, -u])$ is compact, so for each $i \leq 0$, $\gamma_i(u) \in K$ is bounded. Also, the γ_i's are equicontinuous by the same natural parametrization. Therefore, by the Arzéla–Ascoli theorem (See e.g. [7]), our rewritten sequence of unfolding rays $(\gamma_i)_{i\geq 0}$ has a subsequence converging to a ray $\gamma : (-\infty, 0] \to \overline{CV_n}$ with $\gamma(0) = S$. We note that as $\gamma_i(t_i) = T_i' \to T'$ as $i \to -\infty$, we have that the accumulation set of γ contains T'. The following lemma promotes this unfolding path to an unfolding sequence. □

Lemma 1.15. *Let G_t for $t \in (-\infty, 0]$ be an unfolding path whose legal lamination, after taking its diagonal closure, is dual to an arational tree (or, equivalently, the projection of G_t to FF_n converges to a point in ∂FF_n). Then, there is a reduced unfolding sequence whose legal lamination has the same diagonal closure as that of G_t.*

Proof. This is a variant of the proof of Lemma 1.11. First note that the volume and the injectivity radius of G_t go to ∞ as $t \to -\infty$ (or else the projection to FF_n would eventually be constant). We again consider two cases.

The first case is that for arbitrarily large $-t$, the graph G_t contains a rigid vertex. Fix some t. Then there is $t' < t$ so that every noncontractible subgraph of $G_{t'}$ maps onto all of G_t (if this fails for all t', then a truncation of the path would be coarsely constant in FF_n). Consider all points in $G_{t'}$ that map to a rigid vertex in G_t. By the choice of t', all complementary components of this set are trees. Now fold each illegal turn in each component until it either becomes legal and turns into a rigid vertex, or it reaches the boundary of the component, i.e., one of the points that maps to a rigid vertex in G_t. Thus we factored $G_{t'} \to G_t$ as $G_{t'} \to H_t \to G_t$ with the second map sending all vertices to rigid vertices. The graphs H_t obtained in this way when $t \to -\infty$ will form an unfolding sequence with the same legal lamination.

The second case is that after truncation there are no rigid vertices in any G_t. We choose $v_t \in G_t$ for every $t \leq 0$ so that for $t' < t$, $G_{t'} \to G_t$ maps $v_{t'}$ to v_t.

Then, we can pass to a subsequence t_i with $t_i \to -\infty$ such that each v_{t_i} is a vertex of G_{t_i} (otherwise a truncation is coarsely constant in FF_n). As above, after a further subsequence, we may assume that the preimage of v_{t_i} in $G_{t_{i-1}}$ has contractible complementary components. Folding illegal turns within each component produces a factorization $G_{t_{i-1}} \to H \to G_{t_i}$ with all vertices of H mapping to v_{t_i}, and we get an unfolding sequence as above.

It remains to arrange that the unfolding sequence, which we rename G_n, is reduced. Here this is not automatic as in the folding sequence case. However, any proper invariant subgraphs $H_n \subset G_n$ must be forests for large $-n$, since otherwise a truncated sequence would be coarsely constant in FF_n. After truncating we may assume all H_n are forests. Let G'_n be obtained from G_n by collapsing each component of H_n. There are induced maps $G'_{n-1} \to G'_n$. The new sequence G'_n is a reduced unfolding sequence, see [16, Lemma 4.1]. Moreover, the new sequence fellow travels the old in FS_n and hence in FF_n, so it converges to the same point in ∂FF_n per Proposition 1.6. □

We also note that it is possible that an unfolding sequence accumulates on the entire simplex of equivalent arational trees, see [4].

Before we move on the next section, we give a sketch of the proof of Theorem 1.4.

Proof of Theorem 1.4. Given a compatible collection of width measures $\{\mu_n\}_{n=0}^{-\infty}$ on a reduced unfolding sequence, we construct a current μ on the legal lamination Λ of the sequence as follows. For finite directed paths γ and η in G_0, denote by $\langle \gamma, \eta \rangle$ the number of occurrences of γ in η as subwords, where letters are the edges of G_0. Then for each $n \in \mathbb{Z}_{\leq 0}$ and a finite path γ in G_0, we define

$$\alpha_n(\gamma) := \sum_{e_n \in G_n} \mu_n(e_n) \langle \gamma, \phi_{n,0}(e_n) \rangle.$$

It can be shown that the $\alpha_n(\gamma)$ is increasing as $n \to -\infty$, and $\{\alpha_n(\gamma)\}_{n \in \mathbb{Z}_{\leq 0}}$ is bounded above. Hence, for each finite path γ in G_0, the quantity $\mu(\gamma) :=$

$\lim_{n\to-\infty} \alpha_n(\gamma)$ is well defined. Then the set of numbers

$$\{\mu(\gamma) \mid \gamma \text{ is a finite path in } G_0\}$$

satisfies the following: For each finite path γ in G_0 and $\gamma_1 \dots \gamma_k$ are all possible extensions of γ by an edge at the end of γ,

$$\mu(\gamma) = \sum_{i=1}^{k} \mu(\gamma_i).$$

By [23, Lemma 7] (called "Kolmogorov extension property" in [26, Section 2.2]), it follows μ is a current on $\pi_1(G_0) = F_n$. Finally, to show μ is supported on Λ, let η be a leaf in the support of μ. This means for every finite subpath γ of η, we have $\mu(\gamma) > 0$. By definition of μ, this implies for every m with sufficiently large $|m|$, there exists an edge e'_m of G_m such that $\langle \gamma, \phi_{m,0}(e'_m)\rangle > 0$. Since $\phi_{m,0}(e'_m) = \phi_{n,0}(\phi_{m,n}(e'_m))$ is a concatenation of the $\phi_{n,0}$-image of an edge of G_n, taking γ to be longer than the twice of the simplicial length of $\phi_{n,0}(e_n)$ for every $e_n \in G_n$, it follows that $\phi_{n,0}(e_n)$ must be a subpath of γ for some e_n of G_n, so it is a subpath of η. This implies that η is a limit of $\phi_{n,0}(e_n)$ for a sequence of edges e_n in G_n, so η is contained in Λ, concluding the proof. □

1.6. **Transverse decomposition from ergodic measures.** In this section, we present the Transverse Decomposition theorem for (un)folding sequences. This decomposition of a graph into subgraphs reflects the simplex of measures on the corresponding limiting objects. It was first proved by Namazi–Pettet–Reynolds [26], and it was motivated by the corresponding McMullen's theorem [25] in the setting of Teichmüller geodesics. We give an alternative proof of the Transverse Decomposition theorem using only linear algebraic arguments. This is a vital ingredient toward establishing the upper bound in Theorem 4.1.

Recall that for the limiting tree T of a reduced folding sequence and the legal lamination Λ of a reduced unfolding sequence, we denote by $\mathcal{L}(T)$ and by $\mathcal{C}(\Lambda)$ the set of length measures on T and the set of currents on Λ, respectively. We then denote by $\mathbb{P}\mathcal{L}(T)$ and $\mathbb{P}\mathcal{C}(\Lambda)$ the projectivizations of these spaces. These projectivized spaces can be thought of concretely as spaces of length measures or currents, normalized so that a particular conjugacy class has length 1 or compact set has measure 1, respectively. In this way, $\mathbb{P}\mathcal{L}(T)$ and $\mathbb{P}\mathcal{C}(\Lambda)$ become compact, convex sets in a locally convex topological vector space, and we call their extremal points **ergodic**. If $\mathbb{P}\mathcal{L}(T)$ or $\mathbb{P}\mathcal{C}(\Lambda)$ is a singleton, then we call T or Λ, respectively, **uniquely ergodic** and otherwise call T or Λ **nonuniquely ergodic**.

A **transverse decomposition** of a graph G is a collection $H^0, H^1, \dots, H^k$ of subgraphs of G such that the edge set of G is the *disjoint union* of the edge sets of $H^0, H^1, \dots, H^k$. We denote the decomposition by $G = H^0 \sqcup H^1 \sqcup \dots \sqcup H^k$. Note that the vertices may overlap between the subgraphs of the transverse decomposition.

To state the transverse decomposition results, we make some notations as follows. Given a folding/unfolding sequence $(G_s)_s$, we pass to a subsequence so that all G_s are isomorphic graphs and we take a fixed isomorphism from $G = G_0$ to G_s. Denote by e_s the edge of G_s identified to e in G by the isomorphism. Then, for a length measure ℓ on the limiting tree T of a folding sequence $(G_s)_{s\ge 0}$, we denote by ℓ_s the induced length measure on G_s under the isomorphism $\mathcal{L}(T) \cong \mathcal{L}((G_s)_{s\ge 0})$ given in Theorem 1.3. Also, for $e_s \in G_s$, we write $\ell(e_s) := \ell_s(e_s)$. Similarly, for a current μ on the legal lamination Λ of an unfolding sequence $(G_s)_{s\le 0}$, we denote by

μ_s the induced current on G_s under the isomorphism $\mathcal{C}(\Lambda) \cong \mathcal{C}((G_s)_{s\leq 0})$ given in Theorem 1.4. We also write $\mu(e_s) := \mu_s(e_s)$ for $e_s \in G_s$.

Now, we state the Transverse Decomposition theorems and give another proof of the existence of such a decomposition. The original proofs are given in [26, Theorems 4.7, 5.6].

Theorem 1.16 (Transverse Decomposition theorem for a Folding sequence)**.** *Let $(G_s)_{s\geq 0}$ be a reduced folding sequence with limiting tree T and let $\ell^1, \dots, \ell^k$ be the collection of all the distinct (up to scale) ergodic length measures supported on T. Let $(\mu_s)_{s\geq 0}$ be the* frequency current *on the folding sequence, namely $\mu_0(e) = 1$ for every edge e of G_0. Then, after passing to a subsequence, there is a transverse decomposition*

$$G_0 = H^0 \sqcup H^1 \sqcup \dots \sqcup H^k,$$

and

$$H^i := \{e \in EG_0 \mid \lim_{s\to\infty} \mu(e_s)\ell^i(e_s) > 0\}, \text{ for } i = 1, \dots, k, \text{ and}$$
$$H^0 := \{e \in EG_0 \mid \lim_{s\to\infty} \mu(e_s)\ell^i(e_s) = 0, \text{ for all } i = 1, \dots, k\}.$$

Theorem 1.17 (Transverse Decomposition theorem for an Unfolding sequence)**.** *Let $(G_s)_{s\leq 0}$ be a reduced unfolding sequence with legal lamination Λ and let $\mu^1, \dots, \mu^k$ be the collection of all distinct (up to scale) ergodic probability currents supported on Λ. Let $(\ell_s)_{s\leq 0}$ be the* simplicial length measure *on the unfolding sequence, namely $\ell_0(e) = 1$ for every edge e of G_0. Then, after passing to a subsequence, there is a transverse decomposition*

$$G_0 = H^0 \sqcup H^1 \sqcup \dots \sqcup H^k,$$

where

$$H^i := \{e \in EG_0 \mid \lim_{s\to-\infty} \ell(e_s)\mu^i(e_s) > 0\}, \text{ for } i = 1, \dots, k, \text{ and}$$
$$H^0 := \{e \in EG_0 \mid \lim_{s\to-\infty} \ell(e_s)\mu^i(e_s) = 0, \text{ for all } i = 1, \dots, k\}.$$

We say an edge $e \in H^i$ in the transverse decomposition of G_0 for a folding (or unfolding) sequence with $i > 0$ **supports** the ergodic length measure ℓ^i (or the ergodic probability current μ^i, respectively.) This term will be used in Section 4.

We start by giving a proof for the unfolding sequence Theorem 1.17, and afterwards indicate the small changes needed in the proof for the folding sequence Theorem 1.16.

The setup is as follows. Suppose we are given an unfolding sequence, $(G_s)_{s\leq 0}$, where each $G_i \cong G$ and length functions are given by pulling back the simplicial length function ℓ_0 on G_0. Let V be a vector space with basis $\{e^1, \dots, e^n\} = EG$. To each G_s we assign the vector space $V_s \cong V$, but we put a norm on V_s with basis $\{e_s^1, \dots, e_s^n\} = EG_s$, such that $\|e_s^i\|_{V_s} := \ell_s(e_s^i)$ and extend it to an ℓ^1-norm on V_s.

A nonnegative linear combination $a_1 e_s^1 + \dots + a_n e_s^n$ with $a_i \geq 0$ can be interpreted as an assignment of thicknesses to edges, which can then be viewed as rectangles, and then $a_1\|e_s^1\|_{V_s} + \dots + a_n\|e_s^n\|_{V_s}$ is the total area of the graph G_s. In other words, an element μ in the nonnegative cone $V_s^{\geq 0}$ of V_s is an assignment of thicknesses on EG_s, and $\|\mu\|_{\ell^1}$ is the area, used to define the H^i's in the transverse decomposition.

Transition maps on the unfolding sequence $(G_s)_{s\leq 0}$ induce linear maps

$$\dots \longrightarrow V_{-2} \longrightarrow V_{-1} \longrightarrow V_0$$

that are norm nonincreasing, and on the positive cones they are norm (and area) preserving. Hence, we can restrict to the simplex $\Delta_s := \{\mu \in V_s^{\geq 0} \mid \|\mu\|_{\ell^1} = 1\} \subset V_s$ to obtain affine maps

$$\ldots \longrightarrow \Delta_{-2} \longrightarrow \Delta_{-1} \longrightarrow \Delta_0.$$

Denote these affine maps and their compositions by $S_{r,s} : \Delta_r \to \Delta_s$ for $r < s \leq 0$, so that $S_{s,t} \circ S_{r,s} = S_{r,t}$ for $r < s < t \leq 0$. Since all V_s's are isomorphic, it follows that all Δ_s's are isomorphic. Identify all V_s's with V and Δ_s with a fixed $\Delta \subset V$. Note that each $S_{r,s}$ is 1-Lipschitz. Hence by Arzéla–Ascoli, after taking a subsequence, we have the limits

$$S_s := \lim_{r \to -\infty} S_{r,s} : \Delta \to \Delta,$$
$$S := \lim_{s \to -\infty} S_s : \Delta \to \Delta.$$

Lemma 1.18. *The limit $S : \Delta \to \Delta$ is a retraction. That is, $S^2 = S$.*

Proof. From the compatibility condition $S_{s,t} \circ S_{r,s} = S_{r,t}$, we pass to the limit $r \to -\infty$ to get

$$S_{s,t} \circ S_s = S_t.$$

Now take the limit $s \to -\infty$ to get

$$S_t \circ S = S_t.$$

Finally, taking $t \to -\infty$ gives the desired identity $S^2 = S$. □

Corollary 1.19. *The image of S is a simplex. Moreover, Δ can be written as a join of disjoint faces $\Delta^0, \ldots, \Delta^k$ of Δ:*

$$\Delta = \Delta^0 * \Delta^1 * \ldots * \Delta^k,$$

such that the image of S is the convex hull of a finite set that consists of exactly one point $q_i \in \mathrm{int}(\Delta^i)$ for $i = 1, \ldots, k$.

Proof. Since $S^2 = S$, the points in the image of S are fixed. Now take a segment $[x, y] \subset \Delta$ with $x \neq y \in \mathrm{Im}(S)$. Then since V is a vector space, there is a unique line $l \subset V$ containing $[x, y]$. As S is an affine map, it then follows that S fixes $l \cap \Delta$. Hence, $l \cap \Delta \subset \mathrm{Im}(S)$. Therefore, we can consider the smallest affine plane H in V that contains $\mathrm{Im}(S)$. This gives $\mathrm{Im}(S) = H \cap \Delta$.

Since $\mathrm{Im}(S)$ is an affine image of a simplex, it is the convex hull of finitely many extremal points. Let $q_1, \ldots, q_k$ be the vertices of $\mathrm{Im}(S) \subset \Delta$. Let Δ^i be the unique smallest face of Δ that contains q_i. By the previous paragraph, the intersection $\Delta^i \cap \mathrm{Im}(S)$ is precisely $\{q_i\}$. From the fact that q_i is fixed and that S is affine, we can see that S maps Δ^i to q_i. Indeed, if we write q_i as the convex combination of the vertices v_k of Δ^i, $q_i = \sum a_k v_k$, then $a_k > 0$ and $q_i = S(q_i) = \sum a_k S(v_k)$. This shows that $S(v_k)$ is a convex combination of vertices in Δ^i, so we must have that $S(v_k) = q_i$ for each vertex v_k of Δ^i.

In particular, this implies that Δ^i and Δ^j for $i \neq j$ do not have common vertices. Now let Δ^0 be the span of all vertices of Δ that are *not* in any of $\Delta^1, \ldots, \Delta^k$. This finishes the proof. □

We note that the vertices q_i in the above proof represent ergodic measures.

It remains to relate this linear algebra with the statement of the Transverse Decomposition theorem for an unfolding sequence.

Proof of Theorem 1.17. Let $C_s \subset \Delta$ denote the image of S_s. The maps $S_{r,s}$ restrict to surjective affine maps $C_r \to C_s$, so $\dim C_r \geq \dim C_s$ for $r \leq s$. Hence, $\dim C_r \leq \dim \Delta$ stabilizes for sufficiently large $-r$.

However, if $C_s \to C_{s+1}$ is affine, onto, and $\dim C_s = \dim C_{s+1}$, then the map $S_{s,s+1}$ is also one to one. Hence, eventually all C_s are affinely isomorphic. By taking limits, we also get that $\mathrm{Im}(S)$ is naturally affinely isomorphic to each C_s for sufficiently large $-s$. Indeed, we have that $S_t \circ S = S_t$, so $S_t : \mathrm{Im}(S) \to \mathrm{Im}(S_t)$ is surjective, implying for large $-t$, this map is an isomorphism.

For $i = 0, \dots, k$ we will define H^i to be the subgraph whose edges consist of the vertices of Δ^i. See Figure 1. Since the image of S is contained in $\Delta^1 * \dots * \Delta^k$ every column of S has zero e-coordinate for $e \in \Delta^0$. Let $e \in \Delta^0$. Since the e-coordinates of S_s have the form $\ell(e_s)\mu^j(e_s)$ and $\lim S_s = S$, we deduce $\lim_{s\to -\infty} \ell(e_s)\mu^j(e_s) = 0$ for each j, justifying $e \in H^0$. Similarly, H^i for $i \neq 0$ is simply the set of vertices of Δ^i. This means for $e \in H^i$, the e-coordinate of $S(e)$ is nonzero, and this similarly implies $\lim_{s\to -\infty} \ell(e_s)\mu^i(e_s) \neq 0$. Likewise, if $j > 0$ and $j \neq i$, and if $e' \in H^j$, then the e-coordinate of $S(e')$ is 0, so $\lim_{s\to -\infty} \ell(e_S)\mu^j(e_s) = 0$. This concludes the proof. $\square$

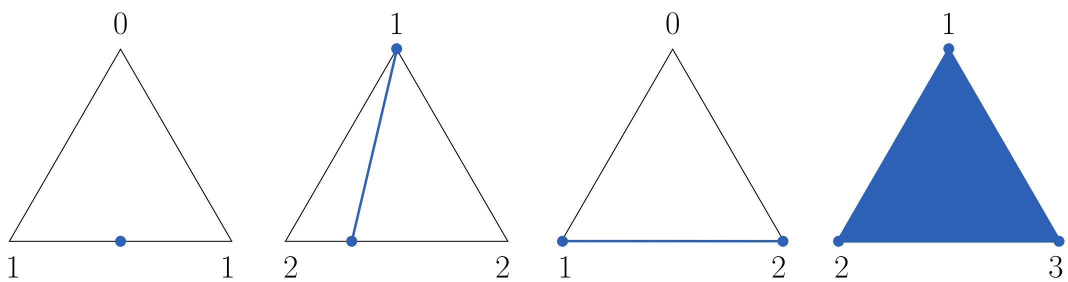

FIGURE 1. The image of S (in blue) and how the vertices of Δ, which correspond to the edges of G, are mapped via S. The vertices labeled by i correspond to the edges of the component H^i in the transverse decomposition. The map S is the retraction that maps Δ onto the simplex $\mathrm{Im}(S)$.

Proof of Theorem 1.16. This time, we give a norm on the basis EG for the vector space V_s associated with G_s using the frequency current, and interpret a nonnegative linear combination in V_s as assigning lengths to edges by their coefficients. We then run the same argument by using that the *transpose* of the transition matrices induce the linear inverse system $\dots \to V_2 \to V_1 \to V_0$, which further reduces to an affine inverse system $\dots \to \Delta_2 \to \Delta_1 \to \Delta_0$. (Recall the compatibility condition for the length measures on the folding sequence in Section 1.3 uses the *transpose*; length measures pull back.) For $s > r$, let $S_{s,r} : \Delta_s \to \Delta_r$ and identify $\Delta_r \cong \Delta$ for all $r \geq 0$. Then define S_r and S as before; $S_r := \lim_{s\to\infty} S_{s,r}$ and $S := \lim_{r\to\infty} S_r$, after taking subsequences. Then S is a retraction. Since the folding sequence is reduced, the inverse limit of the image S_r exists, and it is exactly the space of length measures on the folding sequence. We now follow the same proof of Corollary 1.19 to establish the desired decomposition. $\square$

For the proof of the upper bound given later, we mark here an observation that appeared in the proofs of Theorems 1.16 and 1.17. For folding/unfolding sequences, for $r < s$, the transition map $G_r \to G_s$ induces a linear map between the associated vector spaces V_r and V_s. For an unfolding sequence, it is $V_r \to V_s$ and for a folding sequence, it is $V_s \to V_r$. With normalized bases, denote by $\widetilde{M}^{r,s}$ the transition matrix associated to the linear map induced from $G_r \to G_s$. Denoting by $M^{r,s}$ the transition matrix for the transition map $G_r \to G_s$, the two transition matrices are related as follows.

Lemma 1.20. *For the unfolding sequence, we have*

$$\widetilde{M}^{r,s} = \operatorname{diag}\left(\ell_s(e^1),\ \cdots,\ \ell_s(e^n)\right) M^{r,s} \operatorname{diag}\left(\frac{1}{\ell_r(e^1)},\ \cdots,\ \frac{1}{\ell_r(e^n)}\right).$$

For the folding sequence,

$$\widetilde{M}^{r,s} = \operatorname{diag}\left(\mu_r(e^1),\ \cdots,\ \mu_r(e^n)\right) (M^{r,s})^T \operatorname{diag}\left(\frac{1}{\mu_s(e^1)},\ \cdots,\ \frac{1}{\mu_s(e^n)}\right).$$

Proof. This is because the bases for V_s are normalized by the simplicial length for the unfolding sequence, and by the frequency current for the folding sequence. □

Now, the transverse decomposition results tell that up to relabeling the edges, $\widetilde{M}^{r,s}$ is almost block diagonal with respect to the transverse decomposition.

Corollary 1.21. *Relabel the edges* $e^1, \ldots, e^n$ *of* G *so that for the transverse decomposition* $EG = H^0 \sqcup H^1 \sqcup \ldots \sqcup H^k$, *each* H^i *consists of edges labeled by consecutive integers. Then the transition matrix corresponding to* S *from Lemma 1.18 equals*

$$\widetilde{M} := \lim_{s\to-\infty} \lim_{\substack{r<s \\ r\to-\infty}} \widetilde{M}^{r,s}$$

and it has a block diagonal submatrix with positive matrices corresponding to the edges in $G \setminus H^0$. *A similar statement holds for folding sequences.*

Proof. This follows from the fact that S is a retraction and $\operatorname{Im}(S)$ is a simplex, whose vertices are linear combinations of a disjoint set of vertices of Δ. Each face Δ^i of Δ in Corollary 1.19 forms a diagonal block for $\widetilde{M}$. Since each vertex corresponding to H^i with $i > 0$ is a positive linear combination of vertices in H^i, the diagonal blocks for $H^1, \ldots, H^k$ are positive matrices. □

2. Lower Bound $2n-6$: Folding sequence

In this section, we give a lower bound for the maximum number of linearly independent ergodic measures supported on the limiting tree of a given folding sequence of graphs of rank n.

2.1. **Diagonal-dominant matrices for folding sequences.**

Let n be a positive integer. Here we consider a sequence of $n \times n$ matrices with diagonal entries being dominant over other off-diagonal entries, such that any consecutive product remains diagonal-dominant.

Definition 2.1. Let $\varepsilon > 0$, and let m be a positive integer such that $m \le n$. An $n \times n$ nonnegative matrix $A = (a_{ij})$ with positive diagonal entries is called an (m, ε)**-matrix** if the following holds:

$$\frac{a_{ij}}{a_{jj}} < \varepsilon, \qquad \text{for } j \le m \text{ and } i \neq j.$$

In Section 2, we will primarily be considering (m, ε)-matrices with small ε's converging to zero, and will refer to these matrices as *diagonal-dominant* matrices. We will refer to the first m columns of A as **Tier 1** columns and will call the remaining columns **Tier 2** columns. Then for an $n \times n$ matrix $A = (a_{ij})$, denote by K_A the upper bound of the ratios of any entry to any diagonal entry in a Tier 1 column. Namely,

$$K_A := \max_{\substack{i,j \leq n \\ k \leq m}} \frac{a_{ij}}{a_{kk}}.$$

Given a diagonal entry in a Tier 1 column, the ratio of the other entries of A by this entry is bounded according to the following schematic in Figure 2.

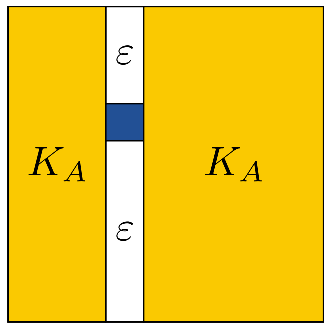

FIGURE 2. The blue cell represents a given diagonal entry in a Tier 1 column. Then each value in the other cells shows an upper bound of the ratio of the entry in that cell to the given diagonal entry.

Lemma 2.2. *Let $n \geq 1$, and fix $1 \leq m \leq n$. Suppose A is an $n \times n$, (m, ε_A)-matrix, and let K_A be as defined above. If A' is an $n \times n$, $(m, \varepsilon_{A'})$-matrix, then there exists $\varepsilon_{AA'} > 0$ such that AA' is an $(m, \varepsilon_{AA'})$-matrix, with*

$$\varepsilon_{AA'} = \varepsilon_A + nK_A\varepsilon_{A'}.$$

Proof. Let A, A', K_A, $\varepsilon_A, \varepsilon_{A'}$, and m be as given in the statement. Then, AA' is a nonnegative matrix and its diagonal entries are positive. Indeed, $(AA')_{ii} = \sum_{j=1}^n a_{ij}a'_{ji} \geq a_{ii}a'_{ii} > 0$. Thus, there exists some $\varepsilon_{AA'} > 0$ such that AA' is an $(m, \varepsilon_{AA'})$-matrix. To bound $\varepsilon_{AA'}$, note for $j \leq m$ and $i \neq j$:

$$\frac{(AA')_{ij}}{(AA')_{jj}} \leq \frac{\sum_k a_{ik}a'_{kj}}{a_{jj}a'_{jj}} = \sum_{k=1}^n \frac{a_{ik}}{a_{jj}} \frac{a'_{kj}}{a'_{jj}}.$$

We now consider bounds on the terms in this summand depending on the value of k.

- ($\mathbf{k = j}$) Here the summand is equal to $\frac{a_{ij}}{a_{jj}}$, which is bounded above by ε_A.
- ($\mathbf{k \neq j}$) We have $\frac{a_{ik}}{a_{jj}} \leq K_A$ and $\frac{a'_{kj}}{a'_{jj}} < \varepsilon_{A'}$, so the summand is bounded above by $K_A\varepsilon_{A'}$.

All in all,

$$\frac{(AA')_{ij}}{(AA')_{jj}} \leq \sum_k \frac{a_{ik}}{a_{jj}} \frac{a'_{kj}}{a'_{jj}} < \varepsilon_A + (n-1)K_A\varepsilon_{A'} \leq \varepsilon_A + nK_A\varepsilon_{A'},$$

so we may take $\varepsilon_{AA'} = \varepsilon_A + nK_A\varepsilon_{A'}$, concluding the proof. $\square$

2.2. **The game: Constructing matrices with increasingly dominant diagonal entries.** In this section, we introduce a game between Alice and Bob which will involve constructing $n \times n$, (m, ε)-matrices with increasingly dominant Tier 1 diagonal entries. Similar constructions were used classically, see e.g. [29]. The purpose of this game, which will henceforth be called the *Alice-Bob game*, is to construct a folding sequence whose transition matrices will satisfy the conditions in Theorem 2.4.

Fix $n \geq 2$ and $1 \leq m \leq n$. The rules of the game are as follows.

First, Alice chooses any $\varepsilon_0 > 0$, and Bob chooses an $n \times n$, (m, ε_0)-matrix A_0. Next, Alice chooses $\varepsilon_1 > 0$, and Bob chooses an $n \times n$, (m, ε_1)-matrix A_1. Continuing inductively, they construct a sequence of $n \times n$ matrices $\{A_j\}_{j=0}^{\infty}$. For $r, s \in \mathbb{Z}_{\geq 0}$ with $r < s$, put $A_{r,s} = A_r A_{r+1} \cdots A_s$. Also, say $A_{r,r} := A_r$. Alice wins the game if there exist $\{\overline{\varepsilon}_r\}_{r \geq 0}$ such that

(1) For every $r \geq 0$ and for each $s > r$, $A_{r,s}$ is an $(m, \overline{\varepsilon}_r)$-matrix, and
(2) $\overline{\varepsilon}_r \to 0$ as $r \to \infty$.

List 1: Conditions on $\{\overline{\varepsilon}_r\}_{r \geq 0}$ for Alice to win the Alice-Bob game.

Otherwise, Bob wins the game.

Theorem 2.3. *Alice has a winning strategy.*

Proof. We will prove that Alice can win the game with $\overline{\varepsilon}_r = 2^{-r}$ for each $r \geq 0$. For ease of exposition, we will denote $\varepsilon_{A_{r,s}}$ by $\varepsilon_{r,s}$ and $K_{A_{r,s}}$ by $K_{r,s}$. Note then $\varepsilon_r = \varepsilon_{r,r}$ and $K_{r,r} = K_{A_r}$. To that end, let Alice choose $\varepsilon_0 = 2^{-1}$, and for each subsequent $s > 0$, choose ε_s so that

$$nP_s \varepsilon_s < 2^{-(s+1)},$$

where $P_s = \max\{K_{r,p} \mid 0 \leq r \leq p < s\}$. Note, in particular, that for each ε_s, we have that $\varepsilon_s \leq 2^{-(s+1)}$.

We claim that for Alice's choices of $\{\varepsilon_p\}_{p=0}^{s}$ as above, $A_{r,s}$ is an $(m, 2^{-r})$-matrix. To see this, note that by Lemma 2.2, we have for each $r \leq p < s$:

$$\varepsilon_{r,p+1} = \varepsilon_{r,p} + nK_{r,p}\varepsilon_{p+1} < \varepsilon_{r,p} + 2^{-(p+2)}.$$

Hence, by telescoping, we see that for each $r \geq 0$ and $s > r$:

$$\begin{aligned}
\varepsilon_{r,s} &< \varepsilon_{r,r} + 2^{-(r+2)} + \ldots + 2^{-s} + 2^{-(s+1)} \\
&\leq 2^{-(r+1)} + 2^{-(r+2)} + \ldots + 2^{-s} + 2^{-(s+1)} \\
&< 2^{-r}.
\end{aligned}$$

Thus, we have shown that for every $r \geq 0$ and for each $s > r$, $A_{r,s}$ is an $(m, \overline{\varepsilon}_r)$ matrix, with $\overline{\varepsilon}_r = 2^{-r}$, since $2^{-r} \to 0$ as $r \to \infty$, we have that Alice has a winning strategy; our choice of $\{\overline{\varepsilon}_r\}_{r \geq 0}$ satisfies the conditions in List 1. $\square$

2.3. **Constructing ergodic length measures from diagonal-dominant matrices.** In this section, we first prove a sufficient condition for when the limiting $\mathbb{R}$-tree of a folding sequence admits at least m linearly independent ergodic length measures.

Theorem 2.4. *Let $(G_s)_{s\geq 0}$ be a reduced folding sequence with limiting tree T, and let $\mathcal{M} = (M^{r,s})_{0\leq r\leq s}$ be the direct system of $n\times n$ transition matrices associated with the folding sequence $(G_s)_{s\geq 0}$. Let $1 \leq m \leq n$. For each $r \geq 0$, suppose there exists $\bar{\varepsilon}_r > 0$ such that for every $s > r$, $(M^{r,s})^T$ is an $(m, \bar{\varepsilon}_r)$-matrix and $\lim_{r\to\infty} \bar{\varepsilon}_r = 0$. Then the limiting tree T admits at least m linearly independent ergodic length measures.*

Note that whenever Alice wins the Alice-Bob game described above (which Theorem 2.3 guarantees will happen), the resultant folding sequence has transition matrices which satisfy the conditions of the above theorem. As such, we will frequently use the Alice-Bob game to construct a folding sequence whose transition matrices satisfy the conditions of the above theorem.

Proof. Firstly, by Theorem 1.3 it suffices to show that the reduced folding sequence $(G_s)_{s\geq 0}$ admits at least m linearly independent measures. Indeed, $\mathcal{C}((G_s)_{s\geq 0})$ is defined as the simplicial cone of non-negative vectors in the n-dimensional vector space whose vectors are sequences $(\vec{v}_s)_{s\geq 0}$ with $\vec{v}_s \in \mathbb{R}^n$ for every $s \geq 0$ satisfying the compatibility condition

$$\vec{v}_s = M_{s+1}^T \vec{v}_{s+1},$$

for every $s \geq 0$, where $M_{s+1} = M^{s,s+1}$. The projectivization is a simplex, and if we show $\dim \mathcal{C}((G_s)_{s\geq 0}) \geq m$ it will follow that the simplex has at least m vertices representing linearly independent ergodic measures. Denoting by $\mathcal{C}(G_s)$ the space of length measures on G_s, the space $\mathcal{C}((G_s)_{s\geq 0})$ is isomorphic to the following space,

$$\mathcal{C}_0 := \lim_{s\to\infty} \bigcap_{k=0}^{s} (M^{0,k})^T(\mathcal{C}(G_k)).$$

On the other hand, we know $\mathcal{C}(G_s) \cong \mathbb{R}^n_{\geq 0}$ for every $s \geq 0$, so it follows that

$$\mathcal{C}_0 = \lim_{s\to\infty} \bigcap_{k=0}^{s} \left(\text{the nonnegative cone of columns of } (M^{0,k})^T\right).$$

Note that we may assume that $\bar{\varepsilon}_0$ is as small as we like by truncating the sequence and reindexing. We choose this so that any $n \times m$ matrix of the form

$$P := \begin{bmatrix} 1 & * & * & \dots & * \\ * & 1 & * & \dots & * \\ * & * & 1 & \dots & * \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ * & * & * & \cdots & 1 \\ * & * & * & \cdots & * \\ * & * & * & \cdots & * \end{bmatrix},$$

where $*$ denotes an entry in the interval $[0, \bar{\varepsilon}_0]$, will have m linearly independent columns.

Now, we claim that $\mathcal{C}_0$ contains at least m linearly independent vectors, by showing that the first m columns of the matrices in the sequence $\{(M^{0,k})^T\}_{k=0}^{\infty}$ *projectively* converge to columns as above (with scaling separate in each column).

For each $1 \leq i \leq m$ and $k \geq 0$, consider the i-th column of $(M^{0,k})^T$ and divide it by the diagonal entry $(M^{0,k})^T_{ii}$. Denote by C_i^k the resulting column vector. We will show that

$$\left[C_1^k \; C_2^k \; \cdots \; C_m^k\right] \quad \longrightarrow \quad P,$$

as $k \to \infty$, after passing to some subsequence. By construction, the i-th entry of C_i^k is 1, for all $k \geq 0$. The other entries of C_i^k are bounded above by $\overline{\varepsilon}_0$. Hence, for each i we can take a further subsequence of the folding sequence to ensure that C_i^k will converge as $k \to \infty$. By passing to further subsequences for each $i = 1, \dots, m$, we may take a subsequence where each $C_1^k, \dots, C_m^k$ converge and the limit is in the form of P. ☐

2.4. **End game: Our folding sequence.** In this section, for each $n \geq 4$, we construct a folding sequence of graphs of rank n whose transition matrices satisfy the conditions of Theorem 2.4.

Theorem 2.5. *For each $n \geq 4$, there exists a reduced folding sequence $\mathcal{G} = (G_k)_{k \geq 0}$ of graphs of rank n whose limiting tree admits at least $2n - 6$ linearly independent ergodic length measures.*

Proof. Let $n \geq 4$. Consider the graph Γ' of rank $n-1$, defined and labeled as in the left graph of Figure 3. The edges $b_0, \dots, b_{n-3}$ are single-edged loops, and $a_0, \dots, a_{n-3}$ are the edges connecting consecutive b_i's. Now consider a loop c, called a *rein*. Then for each $e \in E\Gamma'$, we denote by Γ_e the graph obtained from Γ' by adjoining c to be incident to the edge e such that (c, e) and $(\overline{c}, e)$ form the only illegal turns in Γ_e. See the right graph of Figure 3 for an example of Γ_{a_0}.

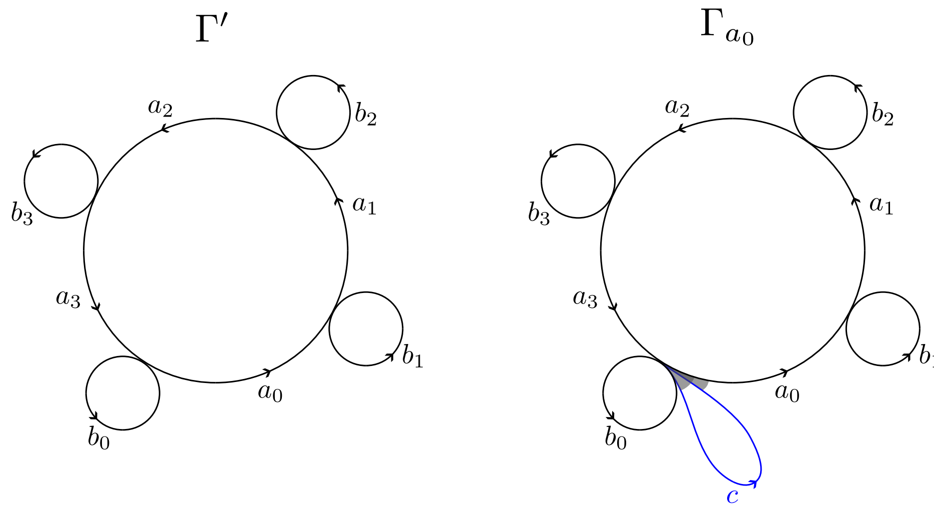

FIGURE 3. The graph Γ' on the left is a graph with $2n - 4$ edges with $n - 2$ vertices, where $n = 6$. The $a_1, \dots, a_{n-3}$ edges together form a single loop in the middle, and the $b_0, \dots, b_{n-3}$ edges are loops incident to each vertex of Γ'. The graph Γ_{a_0} on the right is obtained from Γ' by attaching a loop c that shares the same vertex as the initial vertex of a_0. It has a train track structure with a single gate $\{c, \overline{c}, a_0\}$, which attributes to two illegal turns, shown in gray.

Now, we consider the following sets of maps between different Γ_e's. First, consider the $2n-6$ *rein movers* $\mathcal{R}_{a_0}, \dots, \mathcal{R}_{a_{n-4}}$ and $\mathcal{R}_{b_1}, \dots, \mathcal{R}_{b_{n-3}}$ defined as:

$$\mathcal{R}_{a_i} : \Gamma_{a_i} \to \Gamma_{b_{i+1}}, \qquad \mathcal{R}_{a_i} = \begin{cases} c \mapsto a_i c \overline{a_i}, \\ \mathrm{Id} & \text{otherwise}, \end{cases}$$

$$\mathcal{R}_{b_i} : \Gamma_{b_i} \to \Gamma_{a_i}, \qquad \mathcal{R}_{b_i} = \begin{cases} c \mapsto b_i^2 c \overline{b_i}, \\ \mathrm{Id} & \text{otherwise}. \end{cases}$$

Notice we have b_i^2 in the image of c for $\mathcal{R}_{b_i}$, to prevent c from being an invariant conjugacy class of the composition $\mathcal{F}$ (to be defined below).

Secondly, we fix positive integers $\alpha_1, \dots, \alpha_{n-3}$ and $\beta_1, \dots, \beta_{n-3}$, and consider the $2n-6$ *loopers* $\mathcal{L}_{a_1}, \dots, \mathcal{L}_{a_{n-3}}$ and $\mathcal{L}_{b_1}, \dots, \mathcal{L}_{b_{n-3}}$ defined as:

$$\mathcal{L}_{a_i} : \Gamma_{a_i} \to \Gamma_{a_i}, \qquad \mathcal{L}_{a_i} = \begin{cases} a_i \mapsto c(\overline{a_{i-1}} b_{i-1} a_{i-1} b_i)^{\alpha_i} a_i, \\ \mathrm{Id} & \text{otherwise}, \end{cases}$$

$$\mathcal{L}_{b_i} : \Gamma_{b_i} \to \Gamma_{b_i}, \qquad \mathcal{L}_{b_i} = \begin{cases} b_i \mapsto c \overline{a_{i-1}} b_{i-1}^{\beta_i} a_{i-1} b_i, \\ \mathrm{Id} & \text{otherwise}. \end{cases}$$

Finally, we consider the *rotator* $\rho : \Gamma_{a_{n-3}} \to \Gamma_{a_0}$ defined as

$$\rho = \begin{cases} a_i \mapsto a_{i+1} & \text{for } i = 0, \dots, n-3, \\ b_i \mapsto b_{i+1} & \text{for } i = 0, \dots, n-3, \\ c \mapsto c, \end{cases}$$

where for convenience of notation, we use indices modulo $n-2$, such as $a_{n-2} \equiv a_0$ and $b_{n-2} \equiv b_0$.

Then, our desired train track map $\mathcal{F}$ is the composition of these maps in the following order:

$$\begin{aligned} \mathcal{F} &= \rho \circ (\mathcal{L}_{a_{n-3}} \circ \mathcal{R}_{b_{n-3}}) \circ (\mathcal{L}_{b_{n-3}} \circ \mathcal{R}_{a_{n-4}}) \circ \cdots \circ (\mathcal{L}_{a_1} \circ \mathcal{R}_{b_1}) \circ (\mathcal{L}_{b_1} \circ \mathcal{R}_{a_0}) \\ &= \rho \circ \prod_{i=1}^{n-3} [\mathcal{L}_{a_i} \circ \mathcal{R}_{b_i} \circ \mathcal{L}_{b_i} \circ \mathcal{R}_{a_{i-1}}], \end{aligned}$$

which is a map from Γ_{a_0} to itself.

In fact, we can compute the composition as follows. Each of $a_1, \dots, a_{n-3}$ or of $b_1, \dots, b_{n-3}$ is mapped via the associated looper $\mathcal{L}_{a_i}$ or $\mathcal{L}_{b_j}$ only once in the product $\prod_{i=1}^{n-3} [\mathcal{L}_{a_i} \circ \mathcal{R}_{b_i} \circ \mathcal{L}_{b_i} \circ \mathcal{R}_{a_{i-1}}]$, and only c is altered multiple times via the rein movers. Note that a_0 and b_0 remain the same until the rotator comes in. For brevity, we write:

$$\begin{aligned} \mathcal{P}_1 &:= a_0 b_1 a_1 b_2 a_2 \cdots b_{n-4} a_{n-4} b_{n-3}, \\ \mathcal{P}_2 &:= a_0 b_1^2 a_1 b_2^2 a_2 \cdots b_{n-4}^2 a_{n-4} b_{n-3}^2, \\ \mathcal{Q}_1 &:= a_1 b_2 a_2 b_3 a_3 \cdots b_{n-3} a_{n-3} b_0, \\ \mathcal{Q}_2 &:= a_1 b_2^2 a_2 b_3^2 a_3 \cdots b_{n-3}^2 a_{n-3} b_0^2, \end{aligned}$$

and for a letter e appearing in $\mathcal{P}_1$, write $\mathcal{P}_1[e]$ as the word obtained from truncating the initial part of $\mathcal{P}_1$ up to (but not including) e. For simplicity, for e not appearing in $\mathcal{P}_1$, such as b_0 or a_{n-3}, define $\mathcal{P}_1[e]$ as the empty word. For example, $\mathcal{P}_1[b_2] = b_2 a_2 b_3 a_3 \cdots b_{n-4} a_{n-4} b_{n-3}$. Define $\mathcal{Q}_1[e]$ similarly for a letter e appearing in $\mathcal{Q}_1$, and let $\mathcal{Q}_1[e]$ be the empty word if e is not in $\mathcal{Q}_1$. Define similarly for $\mathcal{P}_2[e]$ and $\mathcal{Q}_2[e]$, for example $\mathcal{Q}_2[b_2] = b_2^2 a_2 b_3^2 a_3 \cdots b_{n-3}^2 a_{n-3} b_0^2$.

Then we can describe the product $\prod_{i=1}^{n-3}[\mathcal{L}_{a_i} \circ \mathcal{R}_{b_i} \circ \mathcal{L}_{b_i} \circ \mathcal{R}_{a_{i-1}}]$ explicitly as follows:

$$\begin{aligned}
c &\longmapsto \mathcal{P}_2 c \overline{\mathcal{P}_1} = a_0 b_1^2 a_1 b_2^2 \cdots b_{n-4}^2 a_{n-4} b_{n-3}^2 c \overline{b_{n-3} a_{n-4} b_{n-4} \cdots b_2 a_1 b_1 a_0},\\
a_0 &\longmapsto a_0,\\
b_0 &\longmapsto b_0,\\
a_k &\longmapsto \mathcal{P}_2[a_k] c \overline{\mathcal{P}_1[a_k]}\, (\overline{a_{k-1}} b_{k-1} a_{k-1} b_k)^{\alpha_k}\, a_k,\\
&= \begin{cases} c\, (\overline{a_{n-4}} b_{n-4} a_{n-4} b_{n-3})^{\alpha_{n-3}}\, a_{n-3} & \text{if } k = n-3,\\ a_k \cdots b_{n-4}^2 a_{n-4} b_{n-3}^2 c \overline{b_{n-3} a_{n-4} b_{n-4} \cdots a_k}\, (\overline{a_{k-1}} b_{k-1} a_{k-1} b_k)^{\alpha_k}\, a_k & \text{if } k = 1, \ldots, n-4, \end{cases}\\
b_k &\longmapsto \mathcal{P}_2[b_k] c \overline{\mathcal{P}_1[b_k]} \left(\overline{a_{k-1}} b_{k-1}^{\beta_k} a_{k-1} b_k\right),\\
&= \begin{cases} b_{n-3}^2 c \overline{b_{n-3}} \left(\overline{a_{n-4}} b_{n-4}^{\beta_{n-3}} a_{n-4} b_{n-3}\right) & \text{if } k = n-3,\\ b_k^2 a_k \cdots b_{n-4}^2 a_{n-4} b_{n-3}^2 c \overline{b_{n-3} a_{n-4} b_{n-4} \cdots a_k b_k} \left(\overline{a_{k-1}} b_{k-1}^{\beta_k} a_{k-1} b_k\right) & \text{if } k = 1, \ldots, n-4. \end{cases}
\end{aligned}$$

Therefore, the image under $\mathcal{F}$ of each edge can be obtained by applying the rotator map ρ to the image above, which has the result of increasing the subscripts of the a_i's and b_i's by 1 where $a_{n-2} \equiv a_0$ and $b_{n-2} \equiv b_0$ while fixing c. Also note for $j = 1, 2$, we have $\rho(\mathcal{P}_j) = \mathcal{Q}_j$, $\rho(\mathcal{P}_j[a_k]) = \mathcal{Q}_j[a_{k+1}]$, and $\rho(\mathcal{P}_j[b_k]) = \mathcal{Q}_j[b_{k+1}]$. Indeed, the $\mathcal{F}$-image of each edge is:

$$\begin{aligned}
c &\longmapsto \mathcal{Q}_2 c \overline{\mathcal{Q}_1} = a_1 b_2^2 a_2 b_3^2 \cdots b_{n-3}^2 a_{n-3} b_0^2 c \overline{b_0 a_{n-3} b_{n-3} \cdots b_3 a_2 b_2 a_1},\\
a_0 &\longmapsto a_1,\\
b_0 &\longmapsto b_1,\\
a_k &\longmapsto \mathcal{Q}_2[a_{k+1}] c \overline{\mathcal{Q}_1[a_{k+1}]}\, (\overline{a_k} b_k a_k b_{k+1})^{\alpha_k}\, a_{k+1},\\
&= \begin{cases} c\, (\overline{a_{n-3}} b_{n-3} a_{n-3} b_0)^{\alpha_{n-3}}\, a_0 & \text{if } k = n-3,\\ a_{k+1} \cdots b_{n-3}^2 a_{n-3} b_0^2 c \overline{b_0 a_{n-3} b_{n-3} \cdots a_{k+1}}\, (\overline{a_k} b_k a_k b_{k+1})^{\alpha_k}\, a_{k+1} & \text{if } k = 1, \ldots, n-4, \end{cases}\\
b_k &\longmapsto \mathcal{Q}_2[b_{k+1}] c \overline{\mathcal{Q}_1[b_{k+1}]} \left(\overline{a_k} b_k^{\beta_k} a_k b_{k+1}\right),\\
&= \begin{cases} b_0^2 c \overline{b_0} \left(\overline{a_{n-3}} b_{n-3}^{\beta_{n-3}} a_{n-3} b_0\right) & \text{if } k = n-3,\\ b_{k+1}^2 a_{k+1} \cdots b_{n-3}^2 a_{n-3} b_0^2 c \overline{b_0 a_{n-3} b_{n-3} \cdots a_{k+1} b_{k+1}} \left(\overline{a_k} b_k^{\beta_k} a_k b_{k+1}\right) & \text{if } k = 1, \ldots, n-4. \end{cases}
\end{aligned}$$

Hence, with $(e^1, \ldots, e^{2n-3}) = (b_1, \ldots, b_{n-3}, a_1, \ldots, a_{n-3}, a_0, b_0, c)$, the transition matrix for $\mathcal{F}$ is:

$$M = \left(\begin{array}{cccccc|cccccc|ccc}
\beta_1 & & & & & & \alpha_1 & & & & & & 0 & 1 & 0 \\
4 & \beta_2 & & & & & \alpha_1 & \alpha_2 & & & & & 0 & 0 & 3 \\
3 & 4 & \beta_3 & & & & 3 & \alpha_2 & \alpha_3 & & & & 0 & 0 & 3 \\
\vdots & \vdots & \vdots & \ddots & & & \vdots & \vdots & \vdots & \ddots & & & \vdots & \vdots & \vdots \\
3 & 3 & 3 & \cdots & \beta_{n-4} & & 3 & 3 & 3 & \cdots & \alpha_{n-4} & & 0 & 0 & 3 \\
3 & 3 & 3 & \cdots & 4 & \beta_{n-3} & 3 & 3 & 3 & \cdots & \alpha_{n-4} & \alpha_{n-3} & 0 & 0 & 3 \\
\hline
2 & & & & & & 2\alpha_1 & & & & & & 1 & 0 & 2 \\
2 & 2 & & & & & 3 & 2\alpha_2 & & & & & 0 & 0 & 2 \\
2 & 2 & 2 & & & & 2 & 3 & 2\alpha_3 & & & & 0 & 0 & 2 \\
\vdots & \vdots & \vdots & \ddots & & & \vdots & \vdots & \vdots & \ddots & & & \vdots & \vdots & \vdots \\
2 & 2 & 2 & \cdots & 2 & & 2 & 2 & 2 & \cdots & 2\alpha_{n-4} & & 0 & 0 & 2 \\
2 & 2 & 2 & \cdots & 2 & 2 & 2 & 2 & 2 & \cdots & 3 & 2\alpha_{n-3} & 0 & 0 & 2 \\
\hline
0 & 0 & 0 & \cdots & 0 & 0 & 0 & 0 & 0 & \cdots & 1 & 1 & 0 & 0 & 0 \\
3 & 3 & 3 & \cdots & 3 & 4 & 3 & 3 & 3 & \cdots & 1 & \alpha_{n-3} & 0 & 0 & 3 \\
1 & 1 & 1 & \cdots & 1 & 1 & 1 & 1 & 1 & \cdots & 1 & 1 & 0 & 0 & 1
\end{array}\right).$$

Hence, the transpose of M is of the following form:

$$M^T = \left(\begin{array}{cccccc|cccccc|ccc}
\beta_1 & 4 & 3 & \cdots & 3 & 3 & 2 & 2 & 2 & \cdots & 2 & 2 & 0 & 3 & 1 \\
 & \beta_2 & 4 & \cdots & 3 & 3 & & 2 & 2 & \cdots & 2 & 2 & 0 & 3 & 1 \\
 & & \beta_3 & \cdots & 3 & 3 & & & 2 & \cdots & 2 & 2 & 0 & 3 & 1 \\
 & & & \ddots & \vdots & \vdots & & & & \ddots & \vdots & \vdots & \vdots & \vdots & \vdots \\
 & & & & \beta_{n-4} & 4 & & & & & 2 & 2 & 0 & 3 & 1 \\
 & & & & & \beta_{n-3} & & & & & & 2 & 0 & 4 & 1 \\
\hline
\alpha_1 & \alpha_2 & 3 & \cdots & 3 & 3 & 2\alpha_1 & 3 & 2 & 2 & \cdots & 2 & 0 & 3 & 1 \\
 & \alpha_2 & \alpha_3 & \cdots & 3 & 3 & & 2\alpha_2 & 3 & 2 & \cdots & 2 & 0 & 3 & 1 \\
 & & \alpha_3 & \cdots & 3 & 3 & & & 2\alpha_3 & 3 & \cdots & 2 & 0 & 3 & 1 \\
 & & & \ddots & \vdots & \vdots & & & \ddots & \ddots & \ddots & \vdots & \vdots & \vdots & \vdots \\
 & & & & \alpha_2 & \alpha_1 & & & & & 2\alpha_{n-4} & 3 & 0 & 3 & 1 \\
 & & & & & \alpha_1 & & & & & & 2\alpha_{n-3} & 1 & \alpha_{n-3} & 1 \\
\hline
0 & 0 & 0 & \cdots & 0 & 0 & 1 & 0 & 0 & \cdots & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 0 & \cdots & 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & 0 & 0 \\
0 & 3 & 3 & \cdots & 3 & 3 & 2 & 2 & 2 & \cdots & 2 & 2 & 0 & 3 & 1
\end{array}\right).$$

Finally, we form a folding sequence $\mathcal{G} = (G_k)_{k\geq 0} = \{\phi_k : G_k \to G_{k+1}\}_{k\geq 0}$ of graphs of rank n with $G_k = \Gamma_{a_0}$ and for $k \geq 0$, let $\phi_k = \mathcal{F}^{(k)}$, which is the map $\mathcal{F}$ with replacing $(\beta_1, \dots, \beta_{n-3}, \alpha_1, \dots, \alpha_{n-3}) = \left(\beta_1^{(k)}, \dots, \beta_{n-3}^{(k)}, \alpha_1^{(k)}, \dots, \alpha_{n-3}^{(k)}\right)$, where the $\alpha_i^{(k)}$ and $\beta_j^{(k)}$ are chosen following Bob's role in Theorem 2.3 with $m = 2n-6$. We note that by construction, in particular by the rotator ρ, the folding sequence does not have a proper invariant subgraph, therefore it is reduced. Now, note that for any $\varepsilon_k > 0$ that Alice chooses, Bob can choose large enough $\alpha_i^{(k)}$ and $\beta_j^{(k)}$ so that the transposed matrix is a $(2n-6, \varepsilon_k)$-matrix. So, Theorem 2.3 guarantees the existence of $\{\overline{\varepsilon_r}\}_{r=0}^{\infty}$ for the assumptions in Theorem 2.4. Therefore, we conclude that the limiting tree admits at least $m = 2n-6$ linearly independent ergodic length measures. $\square$

3. Lower Bound $2n-6$: Unfolding Sequence

In this section, we estimate the lower bound for the number of linearly independent ergodic measures supported on the legal lamination of a given unfolding sequence of graphs of rank n. Theorem 1.4 reduces our problem to finding the number of independent ergodic measures supported on the unfolding sequence.

3.1. **Diagonal-dominant matrices.** Let n be a positive integer. Similar to Section 2.1, here we consider a collection of diagonal-dominant $n \times n$ matrices whose product remains diagonal-dominant. This time however, the definition is a bit more involved. The diagonal entries will be ordered from largest to smallest and we will allow entries below the diagonal to be comparable to the diagonal entry in the same column.

Definition 3.1. Let $p > 1$ and $\varepsilon, \delta > 0$, and let m be a positive integer such that $m \leq n$. An $n \times n$ nonnegative matrix $A = (a_{ij})$ with positive diagonal entries is called an $(m, p, \varepsilon, \delta)$**-matrix** if the following holds:

(a) **(Diagonal entries)** for $i < j \leq m$, we have
$$\frac{a_{jj}}{a_{ii}} < \delta.$$
(b) **(Above diagonal entries)** for $i < j \leq m$, we have
$$\frac{a_{ij}}{a_{jj}} < \varepsilon.$$
(c) **(Below diagonal entries)** for $i < j \leq m$, we have
$$\frac{a_{ji}}{a_{ii}} < p.$$
(d) **(Tier 2 entries)** for $k \leq m < j$ and any i, we have
$$\frac{a_{ij}}{a_{kk}} < p\delta.$$

Recall from Section 2.1 that we refer to the first m columns of the matrix A as **Tier 1** columns, and the other ones as **Tier 2** columns. We will be primarily considering $(m, p, \varepsilon, \delta)$-matrices with small δ and ε and with δ much smaller than ε. We will refer to these matrices as *diagonal-dominant* matrices. Also for an $n \times n$ matrix, denote by K_A the upper bound of the ratios of any entry to any diagonal entry in a Tier 1 column. Namely:
$$K_A := \max_{\substack{i,j \leq n \\ k \leq m}} \frac{a_{ij}}{a_{kk}}.$$

Given a diagonal entry in a Tier 1 column, the ratio of the other entries of A by this entry is bounded according to the following schematic in Figure 4.

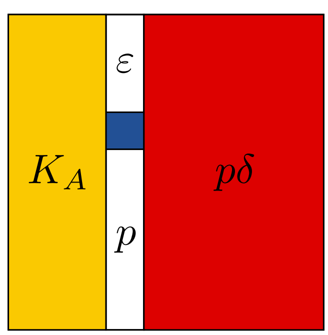

FIGURE 4. The blue cell represents a given diagonal entry in a Tier 1 column. Then each value in the other cells shows an upper bound of the ratio of the entry in that cell to the given diagonal entry. Of course, K_A bounds all such ratios, but p and $p\delta$ are typically much better bounds.

Indeed, the entries above and below the diagonal are controlled by (b) and (c). An entry to the left is controlled by K_A by definition. For entries to the right which are in a Tier 1 column, the ratio bounded by p times the diagonal entry in the same column, which is bounded by δ times the original entry by (a). For the entries to the right which are in a Tier 2 column, the ratio is also bounded by $p\delta$ by (d).

Lemma 3.2. *Let A be an $n \times n$, $(m, p_A, \varepsilon_A, \delta_A)$-matrix with K_A defined as above. If A' is an $n \times n$ $(m, p, \varepsilon, \delta)$-matrix, then there exist $p_{AA'} > 1$, and $\varepsilon_{AA'}, \delta_{AA'} > 0$ such that AA' is an $(m, p_{AA'}, \varepsilon_{AA'}, \delta_{AA'})$-matrix with*

(1) $p_{AA'} = p_A + nK_A\varepsilon + np_A\delta_A p$,
(2) $\varepsilon_{AA'} = \varepsilon_A + nK_A\varepsilon + np_A\delta_A p$, and
(3) $\delta_{AA'} = nK_A p\delta$.

Proof. Let $A = (a_{ij})$ and $A' = (a'_{ij})$ be as given in the statement. Then AA' is a nonnegative matrix with positive diagonal entries. We now check that conditions (a–d) in Definition 3.1 are satisfied for the matrix AA', with $p_{AA'}$, $\varepsilon_{AA'}$, and $\delta_{AA'}$ as desired.

We first establish (a). Let $i < j \le m$. Note that

$$\begin{aligned} \frac{(AA')_{jj}}{(AA')_{ii}} \le \frac{\sum_k a_{jk}a'_{kj}}{a_{ii}a'_{ii}} &= \sum_k \frac{a_{jk}}{a_{ii}}\frac{a'_{kj}}{a'_{ii}} \\ &< \sum_k (K_A)\cdot(p\delta) = nK_A p\delta, \end{aligned} \tag{$*$}$$

so we can take $\delta_{AA'} = nK_A p\delta$.

We now consider (b). Let $i < j \le m$. Similar to above, we see that

$$\frac{(AA')_{ij}}{(AA')_{jj}} \le \frac{\sum_k a_{ik}a'_{kj}}{a_{jj}a'_{jj}} = \sum_k \frac{a_{ik}}{a_{jj}}\frac{a'_{kj}}{a'_{jj}}.$$

This time, we consider the value of each summand depending on the value of k.

- ($\mathbf{k} < \mathbf{j}$) We have $\frac{a_{ik}}{a_{jj}} \le K_A$ and $\frac{a'_{kj}}{a'_{jj}} < \varepsilon$, so each such summand is bounded above by $K_A\varepsilon$.
- ($\mathbf{k} > \mathbf{j}$) We have $\frac{a_{ik}}{a_{jj}} \le p_A\delta_A$ and $\frac{a'_{kj}}{a'_{jj}} < p$, so each such summand is less than $p_A\delta_A p$.
- ($\mathbf{k} = \mathbf{j}$) Here the summand is equal to $\frac{a_{ij}}{a_{jj}}$, which is bounded above by ε_A.

All in all, we have that

$$\frac{(AA')_{ij}}{(AA')_{jj}} \le \sum_k \frac{a_{ik}}{a_{jj}}\frac{a'_{kj}}{a'_{jj}} < \varepsilon_A + nK_A\varepsilon + np_A\delta_A p,$$

so we take $\varepsilon_{AA'} = \varepsilon_A + nK_A\varepsilon + np_A\delta_A p$.

In a similar way, we can prove (c). Let $i < j \le m$, and then

$$\begin{aligned} \frac{(AA')_{ji}}{(AA')_{ii}} \le \frac{\sum_k a_{jk}a'_{ki}}{a_{ii}a'_{ii}} &= \sum_k \frac{a_{jk}}{a_{ii}}\frac{a'_{ki}}{a'_{ii}} \\ &= \left(\sum_{k<i} \frac{a_{jk}}{a_{ii}}\frac{a'_{ki}}{a'_{ii}}\right) + \left(\sum_{k>i} \frac{a_{jk}}{a_{ii}}\frac{a'_{ki}}{a'_{ii}}\right) + \frac{a_{ji}}{a_{ii}} \\ &< nK_A\varepsilon + np_A\delta_A p + p_A, \end{aligned}$$

so we take $p_{AA'} = p_A + nK_A\varepsilon + np_A\delta_A p$.

Finally, we verify (d). Let $t \leq m < j$. Then for any i,

$$\frac{(AA')_{ij}}{(AA')_{tt}} \leq \sum_k \frac{a_{ik}}{a_{tt}} \frac{a'_{kj}}{a'_{tt}} < nK_A p\delta = \delta_{AA'} < p_{AA'}\delta_{AA'}$$

since $p_{AA'} > p_A > 1$. □

3.2. **The game: Constructing matrices with increasingly dominant diagonal entries.** As in Section 2.2, we consider another game by Alice and Bob. Due to the more involved definition of the diagonal dominant matrix used in this section, they engage in a game with more parameters as well. Alice and Bob play the following game. Fix $n \geq 2$, $m \leq n$, and $p > 1$. First, Alice chooses $(\varepsilon_0, \delta_0)$. Then Bob chooses an $n \times n$, $(m, p, \varepsilon_0, \delta_0)$-matrix A_0. Then Alice chooses $(\varepsilon_1, \delta_1)$ and Bob chooses an $n \times n$, $(m, p, \varepsilon_1, \delta_1)$-matrix A_1. Continuing inductively, they construct a sequence of $n \times n$ matrices $\{A_j\}_{j\geq 0}$. For $r < s$, put $A_{r,s} = A_r A_{r+1} \cdots A_s$. Write $A_{r,r} := A_r$. Alice wins the game if there exist triples $\{(\overline{p}_r, \overline{\varepsilon}_r, \overline{\delta}_r)\}_{r\geq 0}$ such that

(1) for each $s > r$, $A_{r,s}$ is an $(m, \overline{p}_r, \overline{\varepsilon}_r, \overline{\delta}_r)$-matrix,
(2) $\{\overline{p}_r\}_{r\geq 0}$ is uniformly bounded,
(3) $\overline{\varepsilon}_r \to 0$ as $r \to \infty$, and
(4) $\overline{\delta}_r \to 0$ as $r \to \infty$.

List 2: Conditions on $\{(\overline{p}_r, \overline{\varepsilon}_r, \overline{\delta}_r)\}_{r\geq 0}$ for Alice to win in the Alice-Bob game.

Otherwise, Bob wins the game.

Theorem 3.3. *Alice has a winning strategy.*

Proof. We will show that Alice can make choices such that conditions (1–4) in List 2 hold. It will be convenient to simplify notation in this proof and denote by $K_{r,s}, p_{r,s}, \varepsilon_{r,s}, \delta_{r,s}$ the corresponding constants for the matrix $A_{r,s}$. In particular, $\varepsilon_{r,r} = \varepsilon_r$ and $\delta_{r,r} = \delta_r$. We also have $p > 1$ without the subscript, chosen before the game starts. For Alice to choose $(\varepsilon_s, \delta_s)$, it will be important to see all matrices A_r for $r < s$. First, we claim the following:

Claim 3.4. *By choosing δ_i sufficiently small at each step, Alice can ensure that*

$$\delta_{r,s} < \frac{1}{2^s},$$

for all $r \geq 0$ and $s > r$.

Proof of the claim. Let Alice choose $\delta_0 = 1$, and for each subsequent $s > 0$, choose δ_s so that

$$nP_s p\delta_s < 2^{-s},$$

where $P_s = \max\{K_{r,t} \mid r \leq t < s\}$. Then by Lemma 3.2 for $A_{r,s} = A_{r,s-1}A_s$, we have

$$\delta_{r,s} \leq nK_{r,s-1}p\delta_s \leq nP_s p\delta_s < 2^{-s} < 2^{-r}$$

holds for each $r \geq 0$ and $s > r$, concluding the claim. △

In light of Claim 3.4, set $\overline{\delta}_r := 2^{-r}$ for $r \geq 0$.

Next, note that the increase in the p-constant and the ε-constant of AA' is bounded by the same amount $nK_A\varepsilon + np_A\delta_A p$. We apply this to $A_{r,s}A_{s+1} = A_{r,s+1}$. Then the corresponding increment to $\varepsilon_{r,s}$ and $p_{r,s}$ is:

$$nK_{r,s}\varepsilon_{s+1} + np_{r,s}\delta_{r,s}p.$$

By choosing small ε_{s+1}, Alice can ensure that the first term is arbitrarily small. Claim 3.4 ensures the second term to be smaller than $\frac{npp_{r,s}}{2^s}$. Hence, we can take the sum of the two terms to be no larger than $\frac{npp_{r,s}}{2^s}$. In other words, by choosing small $(\varepsilon_{s+1}, \delta_{s+1})$, we have for $r \leq s$:

$$p_{r,s+1} \leq p_{r,s} + \frac{npp_{r,s}}{2^s} = p_{r,s}\left(1 + \frac{np}{2^s}\right), \tag{$*_1$}$$

$$\varepsilon_{r,s+1} \leq \varepsilon_{r,s} + \frac{npp_{r,s}}{2^s}. \tag{$*_2$}$$

Multiplicatively telescoping Inequality ($*_1$), and noting $p_{r,r} = p$,

$$p_{r,s+1} \leq p\left(1 + \frac{np}{2^r}\right) \cdots \left(1 + \frac{np}{2^{s-1}}\right)\left(1 + \frac{np}{2^s}\right).$$

Note the infinite product

$$\prod_{k=r}^{\infty}\left(1 + \frac{np}{2^k}\right) \tag{$**$}$$

converges, since the infinite sum $\sum_{k=r}^{\infty} \frac{np}{2^k}$ converges. Take $\overline{p}_r$ to be the infinite product ($**$) multiplied by p. Since $p = p_{r,r} < \overline{p}_r$, it follows that $p_{r,s} \leq \overline{p}_r$ for all $r \geq 0$ and $s \geq r$, and $\{\overline{p}_r\}_{r\geq 0}$ is uniformly bounded above by $\prod_{k=0}^{\infty}\left(1 + \frac{np}{2^k}\right)$

Similarly, picking up from Inequality ($*_2$), we have

$$\varepsilon_{r,s+1} \leq \varepsilon_{r,s} + \frac{npp_{r,s}}{2^s} \leq \varepsilon_{r,s} + \frac{np\overline{p}_r}{2^s},$$

for each $r \geq 0$ and $s \geq r$. By telescoping with a fixed r, and varying s to r,

$$\varepsilon_{r,s+1} \leq \varepsilon_r + \sum_{k=r}^{s} \frac{np\overline{p}_r}{2^k}.$$

Therefore, by letting $\overline{\varepsilon}_r$ be $\overline{\varepsilon}_r := \varepsilon_r + \sum_{k=r}^{\infty} \frac{np\overline{p}_r}{2^k}$, we have that $\varepsilon_{r,s} \leq \overline{\varepsilon}_r$ for every $r \geq 0$ and $s \geq r$, and that $\overline{\varepsilon}_r \to 0$ as $r \to \infty$ (since the infinite sum converges to 0 as $r \to 0$ and Alice can ensure that $\varepsilon_r \to 0$), concluding the proof. □

3.3. Constructing ergodic currents from diagonal-dominant matrices.

Theorem 3.5. *Let $(G_s)_{s\leq 0}$ be a reduced unfolding sequence with legal lamination $\Lambda \subset G_0$, and $\mathcal{M} = (M^{r,s})_{r\leq s\leq 0}$ be the inverse system of $n \times n$ transition matrices associated with the unfolding sequence $(G_s)_{s\leq 0}$. Let $1 \leq m \leq n$. Suppose that for each $s \leq 0$, there exists a positive triple $(\overline{p}_s, \overline{\varepsilon}_s, \overline{\delta}_s)$ such that for every $r < s$, $M^{r,s} \in \mathcal{M}$ is an $(m, \overline{p}_s, \overline{\varepsilon}_s, \overline{\delta}_s)$ matrix such that*

(i) $\{\overline{p}_s\}_{s\leq 0}$ *is uniformly bounded,*
(ii) $\lim_{s\to-\infty} \overline{\varepsilon}_s = 0$, *and*
(iii) $\lim_{s\to-\infty} \overline{\delta}_s = 0$.

Then the legal lamination Λ admits at least m linearly independent ergodic measures.

Proof. The proof of this result mirrors the proof of Theorem 2.4. Firstly, by Theorem 1.4 it suffices to show the reduced unfolding sequence $(G_s)_{s\leq 0}$ admits at least m linearly independent ergodic measures. Recall $\mathcal{C}((G_s)_{s\leq 0})$ is defined as the cone of non-negative vectors in the n-dimensional vector space whose vectors are sequences $(\vec{v}_s)_{s\leq 0}$ with $\vec{v}_s \in \mathbb{R}^n$ for every $s \leq 0$ satisfying the compatibility condition

$$\vec{v}_s = M_{s-1}\vec{v}_{s-1},$$

for every $s \leq 0$, where $M_{s-1} = M^{s-1,s}$. Denoting by $\mathcal{C}(G_s)$ the space of currents on G_s, the space $\mathcal{C}((G_s)_{s\leq 0})$ is isomorphic to the following space,

$$\mathcal{C}_0 := \lim_{r\to-\infty} \bigcap_{k=r}^{0} M^{k,0}(\mathcal{C}(G_k)).$$

On the other hand, we know $\mathcal{C}(G_s) \cong \mathbb{R}^n_{\geq 0}$ for every $s \leq 0$, so it follows that

$$\mathcal{C}_0 = \lim_{r\to-\infty} \bigcap_{k=r}^{0} \left(\text{the nonnegative cone of columns of } M^{k,0}\right).$$

After truncating the sequence and reindexing, we may assume that $\bar{\varepsilon}_0$ is so small that any $n \times m$ matrix of the form

$$P := \begin{bmatrix} 1 & * & * & \dots & * \\ * & 1 & * & \dots & * \\ * & * & 1 & \dots & * \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ * & * & * & \cdots & 1 \\ * & * & * & \cdots & * \\ * & * & * & \cdots & * \end{bmatrix},$$

with entries above the diagonal bounded by $\bar{\varepsilon}_0$ and entries below the diagonal bounded by $\sup \bar{p}_s$ will have m linearly independent columns.

Now we claim that $\mathcal{C}_0$ contains at least m linearly independent vectors, by showing that the first m columns of the matrices in the sequence $\{M^{k,0}\}_{k=-\infty}^{0}$ *projectively* (with separate scaling in each column) converge to m vectors as above. The details are analogous to the proof of Theorem 2.4 and are thus omitted. □

3.4. **End game: Our unfolding sequence.** In this section, for each $n \geq 4$, we construct an unfolding sequence of graphs of rank n whose transition matrices satisfy the conditions of Theorem 3.5.

Theorem 3.6. *For each $n \geq 4$, there exists a reduced unfolding sequence $\mathcal{G} = (G_k)_{k\leq 0}$ of graphs of rank n whose legal lamination admits at least $2n-6$ linearly independent ergodic measures.*

Proof. Following the construction in Theorem 2.5, build the same graph Γ_{a_0} with the maps $\mathcal{F}^{(k)}$, to form a reduced unfolding sequence $\mathcal{G} = (G_k)_{k\leq 0} = \{\phi_{k-1} : G_{k-1} \to G_k\}_{k\leq 0}$ of graphs of rank n with $G_k = \Gamma_{a_0}$ and $\phi_k = \mathcal{F}^{(k)}$. The only difference we will make for our current setting of an unfolding sequence is that we will label the edges of Γ_{a_0} so that

$$(e^1, \dots, e^{2n-3}) = (a_{n-3}, a_{n-2}, \dots, a_1, b_{n-3}, b_{n-2}, \dots, b_1, a_0, b_0, c).$$

Then, our transition matrix M for $\mathcal{F}$ will be of the form:

$$
\left(\begin{array}{cccccc|cccccc|ccc}
2\alpha_{n-3} & 3 & 2 & 2 & \cdots & 2 & 2 & 2 & 2 & 2 & \cdots & 2 & 0 & 0 & 2 \\
 & 2\alpha_{n-4} & 3 & 2 & \cdots & 2 & & 2 & 2 & 2 & \cdots & 2 & 0 & 0 & 2 \\
 & & \ddots & \ddots & \ddots & \vdots & & & \ddots & \ddots & \ddots & \vdots & \vdots & \vdots & \vdots \\
 & & & 2\alpha_3 & 3 & 2 & & & & 2 & 2 & 2 & 0 & 0 & 2 \\
 & & & & 2\alpha_2 & 3 & & & & & 2 & 2 & 0 & 0 & 2 \\
 & & & & & 2\alpha_1 & & & & & & 2 & 1 & 0 & 2 \\
\hline
\alpha_{n-3} & \alpha_{n-4} & 3 & 3 & \cdots & 3 & \beta_{n-3} & 4 & 3 & 3 & \cdots & 3 & 0 & 0 & 3 \\
 & \alpha_{n-4} & \alpha_{n-5} & 3 & \cdots & 3 & & \beta_{n-4} & 4 & 3 & \cdots & 3 & 0 & 0 & 3 \\
 & & \ddots & \ddots & \ddots & \vdots & & & \ddots & \ddots & \ddots & \vdots & \vdots & \vdots & \vdots \\
 & & & \alpha_3 & \alpha_2 & 3 & & & & \beta_3 & 4 & 3 & 0 & 0 & 3 \\
 & & & & \alpha_2 & \alpha_1 & & & & & \beta_2 & 4 & 0 & 0 & 3 \\
 & & & & & \alpha_1 & & & & & & \beta_1 & 0 & 1 & 0 \\
\hline
1 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & 0 \\
\alpha_{n-3} & 3 & 3 & 3 & \cdots & 3 & 4 & 3 & 3 & 3 & \cdots & 3 & 0 & 0 & 3 \\
1 & 1 & 1 & 1 & \cdots & 1 & 1 & 1 & 1 & 1 & \cdots & 1 & 0 & 0 & 1
\end{array}\right)
$$

which can be seen as Bob's $(2n-6, p=2, \varepsilon_r, \delta_r)$-matrix with appropriate choice of $(\alpha_1, \dots, \alpha_{n-3}, \beta_1, \dots, \beta_{n-3})$ for each of Alice's choices of $\varepsilon_r > 0$ and $\delta_r > 0$. Hence, Alice's winning sequence $\{(\varepsilon_r, \delta_r)\}_{r \le 0}$ in Theorem 3.3 satisfying the conditions (i–iii) of Theorem 3.5 provides at least $m = 2n-6$ linearly independent ergodic measures on the legal lamination of our reduced unfolding sequence $\mathcal{G}$. □

4. Upper Bound $2n-1$

In Theorem 1.10 we have seen that for every arational limiting tree T, there exists a reduced folding sequence limiting to T. Likewise, by Theorem 1.14, for every arational legal lamination Λ, there exists an unfolding sequence whose legal lamination has diagonal closure equal to Λ. Hence, to establish the number of ergodic measures on an arational tree or arational legal lamination, it suffices to prove the following.

Theorem 4.1. *Let $(G_s)_{s\ge 0}$ be a reduced folding sequence of graphs of rank n. Then, the number of linearly independent ergodic measures on the limiting tree of the folding sequence is bounded above by $2n-1$. Similarly, given a reduced unfolding sequence $(G_s)_{s\le 0}$ of graphs of rank n, the number of linearly independent ergodic measures on the legal lamination of the unfolding sequence is also bounded above by $2n-1$.*

We will order the edges of $G_0 = G$ so that if we build the graph G by adding one edge at a time in the increasing order, the 'complexity' (Definition 4.6) of the resultant graph increases for each addition. We will then use that the total increase of complexity is bounded above by $-\chi(G) = n-1$.

Intuitively, we give an ordering so that smaller edges are 'easier' to be traversed by the images of other edges. For folding sequences, smaller edges correspond to *wider* edges, and for unfolding sequences, they correspond to *shorter* edges. This correspondence is analogous to the idea that in an electrical current, a wider and shorter resistor has a smaller resistance than a thinner and longer resistor.

To obtain such an ordering from a *folding sequence* $(G_s)_{s\ge 0}$, we consider the **frequency current** μ on G_0, namely $\mu(e^i) = 1$ for $i = 1, \dots, n$. Then, μ pushes forward to μ_s on G_s for $s \ge 0$. We then pass to a subsequence to ensure that for all pairs of edges e, e' in G_0, $\lim_{s\to\infty} \frac{\mu_s(e'_s)}{\mu_s(e_s)}$ exists in $[0, \infty]$. Next, for edges e, e' of G_0, we define:

$$e < e' \quad \text{if} \quad \lim_{s\to\infty} \frac{\mu_s(e'_s)}{\mu_s(e_s)} = 0, \text{ and}$$
$$e \sim e' \quad \text{if} \quad \lim_{s\to\infty} \frac{\mu_s(e'_s)}{\mu_s(e_s)} \in (0,\infty),$$

and write $e \lesssim e'$ if $e < e'$ or $e \sim e'$. Here, we defined the order in a way that $e < e'$ if the width of e grows *faster* than that of e'.

Similarly, for an *unfolding sequence* $(G_s)_{s\le 0}$, let ℓ be the **simplicial length function** on G_0, namely $\ell(e^i) = 1$ for $i = 1, \dots, n$. Then, ℓ pulls back to ℓ_s on G_s for $s \le 0$. We then pass to a subsequence such that for each pair of edges e, e' of G_0, either $\lim_{s\to-\infty} \frac{\ell_s(e_s)}{\ell_s(e'_s)}$ or $\lim_{s\to-\infty} \frac{\ell_s(e'_s)}{\ell_s(e_s)}$ exists. Finally, for edges e, e' in G_0, we define:

$$e < e' \quad \text{if} \quad \lim_{s\to-\infty} \frac{\ell_s(e_s)}{\ell_s(e'_s)} = 0,$$
$$e \sim e' \quad \text{if} \quad \lim_{s\to-\infty} \frac{\ell_s(e_s)}{\ell_s(e'_s)} \in (0,\infty),$$

and denote by $e \lesssim e'$ if $e \sim e'$ or $e < e'$. Thus $e < e'$ if the length of e grows *slower* than that of e'.

In Sections 4.1 and 4.2, we prepare the preliminary lemmas necessary for the proof of Theorem 4.1. In Section 4.3, we utilize those lemmas along with a common combinatorial trick to establish the $2n-1$ upper bound for both unfolding and folding sequences at the same time.

4.1. **Setup for unfolding sequence.** The first step is to exploit the reduced hypothesis on the unfolding sequence. Recall that given finite directed labeled paths γ and ω, we use the notation $\langle \gamma, \omega\rangle$ to denote the number of times γ appears as a (directed) subpath of ω.

Lemma 4.2 (Reduced Unfolding Sequence). *Let $(G_s)_{s\le 0}$ be an unfolding sequence with every graph isomorphic to $G_0 = G$. If $(G_s)_{s\le 0}$ is reduced, then for edges e, e' of G and any number $K > 0$, there exists $N < 0$ such that for each $s < N$, we have that*

$$\langle e'_0, \phi_{s,0}(e_s)\rangle > K.$$

Proof. Let $e_0 \in EG_0$ and $K > 0$. For each $s \le 0$, consider the subgraph H_s of G_s whose edge set is given by

$$EH_s = \{f_s \in EG_s \mid \langle e_0, \phi_{s,0}(f_s)\rangle \le K\}.$$

For each $s < 0$, we claim $\phi_{s,s+1}(H_s) \subset H_{s+1}$. Indeed, pick $f_s \in EH_s$. Because $\phi_{s,0} = \phi_{s+1,0} \circ \phi_{s,s+1}$,

$$\langle e_0, \phi_{s+1,0}(\phi_{s,s+1}(f_s))\rangle \le K.$$

Now for any g_{s+1} contained in $\phi_{s,s+1}(f_s)$ we have

$$\langle e_0, \phi_{s+1,0}(g_{s+1})\rangle \le K,$$

showing $g_{s+1} \in EH_{s+1}$. This establishes the claim and therefore $(H_s)_{s\le 0}$ is an invariant sequence of subgraphs.

As $(G_s)_{s\le 0}$ is reduced, it follows that $(H_s)_{s\le 0}$ is eventually either the sequence of whole graphs or the sequence of empty sets. In the latter case we are done so assume that $H_s = G_s$ for every $s < N$ for some $N < 0$. As the transition maps of the unfolding sequence are surjective, the sequence $(\langle e_0, \phi_{s,0}(G_s)\rangle)_{s<N}$ is

a nondecreasing sequence of positive integers bounded above by $K|EG|$. Hence, we may assume the sequence $(\langle e_0, \phi_{s,0}(G_s)\rangle)_{s<N}$ is constant. This implies that if $e_s' \subset G_s$ is any edge such that $\phi_{s,0}(e_s')$ crosses e_0 and if x_s is a point in the interior of e_s' then the preimages of x_s in G_t for $t < s$ all consist of a single point x_t. But then the sequence K_t of subgraphs of G_t consisting of all edges except the one containing x_t is invariant, contradicting the assumption that the sequence G_t is reduced (G has rank more than one). $\square$

Lemma 4.3 (Witness Lemma for Unfolding Sequence)**.** *Let $(G_s)_{s\leq 0}$ be a reduced unfolding sequence with every graph isomorphic to G. For $e \in EG$ such that $e \in H^i$ for some $i > 0$, there exists an immersed loop α in G such that:*

(1) α contains e,
(2) α does not go through $e' \in EG$ with $e' > e$, and
(3) α does not go through $e' \in EG$ with $e' \sim e$ and $e' \in H^j$ for some $j \neq i$.

Proof. Let $e \in H^i$ for some $i > 0$ and $e' \in EG$ be another edge. For $r < s$, denote by $M := M^{r,s}$ the transition matrix for the transition map $G_r \to G_s$, and denote by $\widetilde{M} := \widetilde{M}^{r,s}$ the transition matrix for the linear map $V_r \to V_s$ induced by the map $G_r \to G_s$, with normalized bases. Up to taking a subsequence, the normalized transition matrices $\widetilde{M}^{r,s}$ converge to the retraction S. Hence, by Corollary 1.21, we have that for large enough $|r|, |s|$ such that $r < s < 0$, the diagonal entry $\widetilde{M}_{ee} := \widetilde{M}^{r,s}_{ee}$ of the normalized matrix $\widetilde{M}^{r,s}$ is bounded away from 0. Moreover, according to Lemma 1.20, we have that

$$(\star) \qquad \frac{\widetilde{M}_{e'e}}{\widetilde{M}_{ee}} = \frac{M_{e'e}\frac{\ell_r(e')}{\ell_s(e)}}{M_{ee}\frac{\ell_r(e)}{\ell_s(e)}} = \frac{M_{e'e}}{M_{ee}} \cdot \frac{\ell_r(e')}{\ell_r(e)}.$$

Since $\widetilde{M}_{ee} > \varepsilon > 0$ and $\widetilde{M}_{e'e}$ is bounded (as $\widetilde{M}^{r,s}$ converges to S), $\frac{\widetilde{M}_{e'e}}{\widetilde{M}_{ee}}$ is bounded above. We divide into two cases with a goal to achieve Property (2) and (3) for a witness loop.

Case 1. $e' > e$. For this case, $e' > e$ implies $\frac{\ell_r(e')}{\ell_r(e)} \to \infty$ as $r \to -\infty$. Therefore, it follows that $\frac{M_{e'e}}{M_{ee}} \to 0$ as $r, s \to -\infty$.

Case 2. $e' \sim e$ **but** $e' \in H^j$ **for some** $j \neq i$. Again by Corollary 1.21, we have

$$\lim_{r\to-\infty} \lim_{\substack{s<r \\ s\to-\infty}} \frac{\widetilde{M}_{e',e}}{\widetilde{M}_{e,e}} = 0.$$

However, as $e \sim e'$, the lengths $\ell_r(e)$ and $\ell_r(e')$ are comparable as $r \to -\infty$, so by Equation ($\star$), it follows that

$$\lim_{r\to-\infty} \lim_{\substack{s<r \\ s\to-\infty}} \frac{M_{e',e}}{M_{e,e}} = 0.$$

All in all, in both cases, the number of e_s' in $\phi_{r,s}(e_r)$ is sublinear in the number of e_s in $\phi_{r,s}(e_r)$ as $r < s$ and $r, s \to -\infty$. Thus, in the immersed path $\phi_{r,s}(e_r)$ in G_s there exist two occurrences of e_s's with the same orientation, say $e_{s,1}Ae_{s,2} \subset \phi_{r,s}(e_r)$, such that in the gap A between them there is no e' such that $e' > e$ or $e' \sim e$ with $e' \in H^j$ for some $j \neq i$. Then $\alpha = e_s A$ is the desired witness loop. $\square$

4.2. **Setup for folding sequence.** Mirroring Section 4.1, we set up lemmas for folding sequences.

Lemma 4.4 (Reduced Folding Sequence). *Let $(G_s)_{s\geq 0}$ be a folding sequence with every graph isomorphic to G. If $(G_s)_{s\geq 0}$ is reduced, then for edges e, e' of G and any number $K > 0$, there exists $N > 0$ such that*

$$\langle e'_s, \phi_{0,s}(e_0)\rangle > K,$$

for each $s > N$.

Proof. Fix $K > 0$ and $s > 0$. Let $e_s \in EG_s$. For $0 \leq i \leq s$, consider the subgraph $H^i_s \subset G_i$ whose edge set is given by:

$$EH^i_s = \{f_i \in EG_i \mid \langle e_s, \phi_{i,s}(f_i)\rangle \leq K\}.$$

We note here $H^s_s = G_s$. Using the same argument as in the proof of Lemma 4.2, we have $\phi_{i,i+1}(H^i_s) \subset H^{i+1}_s$ for $0 \leq i < s$.

Now we want to show that all but finitely many $s > 0$, H^0_s is the empty set, for every choice $e_s \in EG_s$. Suppose for the sake of contradiction that there exist infinitely many $s > 0$ such that H^0_s is not empty for some choice of $e_s \in EG_s$. As G_0 is a finite graph, there could be only finitely many different subgraphs of G_0. Hence, we may take an infinite subsequence of such s's such that H^0_s's are the same nonempty subgraph of G_0. By taking a further subsequence, we may assume the H^0_s's are constructed with e_s sharing the same $e \in EG$. Repeating the same argument, we may assume there exists a sequence $(s_j)_{j>0}$ of positive integers such that for each $i \geq 0$, whenever defined, $H^i_{s_j}$'s are all identical for $j > 0$. Hence, for each $i \geq 0$, set $H_i = H^i_{s_j}$ for any $j > 0$ such that $s_j \geq i$. Then as $\phi_{i,i+1}(H^i_s) \subset H^{i+1}_s$, it follows that $(H_i)_{i\geq 0}$ is an invariant sequence of subgraphs. By the reducibility of $(G_s)_{s>0}$, it follows that $(H_i)_{i\geq 0}$ is eventually either the sequence of empty sets or the sequence of the whole graphs.

Suppose that $(H_i)_{i\geq 0}$ is eventually the sequence of the whole graphs. Namely, say there is $N > 0$ such that $H_i = G_i$ for each $i \geq N$. Then by the identification above, there exists an increasing sequence $(t_j)_{j\geq 1}$ such that $t_1 > N$ and $e_{t_j} \in EG_{t_j}$ such that

$$H_N = \{f_N \in EG_N \mid \langle e_{t_j}, \phi_{N,t_j}(f_N)\rangle \leq K\}$$

for each $j \geq 1$. Since $H_N = G_N$, we have

$$\langle e_{t_j}, \phi_{N,t_j}(G_N)\rangle \leq |EG| \cdot K$$

for each $j \geq 1$. Recall we have reduced to the case where H_N's are constructed with the same $e \in EG$. Then as ϕ_{N,t_j}'s are surjective, it follows that the sequence $(\langle e_{t_j}, \phi_{N,t_j}(G_N)\rangle)_{j\geq 1}$ is a nondecreasing sequence, bounded above by $K|EG|$. Hence it is eventually constant. Say it is constant for $j \geq M$. This means that for $i \geq t_M$, no folding is happening on the preimages of e_{t_j} in G_i, where $j > 0$ is any integer such that $t_j > i$. Now for $i \geq t_M$, take the subgraph H'_i of G_i whose edge set consists of the edges not mapped over e_{t_j} for every j such that $t_j > i$. Then $(H'_i)_{i\geq t_M}$ is a sequence of invariant subgraphs, which cannot be the whole graph as ϕ_{i,t_j} is surjective. Hence by the reducibility of $(G_s)_{s\geq 0}$, $(H'_i)_{i\geq t_M}$ is the sequence of empty graphs. Running the same argument with different choice of e_{t_j} that appears in the identifications of H_N, we conclude that every sequence (H'_i) associated to each choice of e_{t_j} is the sequence of empty sets. However, this implies that eventually there is no folding happening on $(G_s)_{s\geq 0}$, contradiction.

All in all, we conclude that $(H_i)_{i\geq 0}$ is eventually a sequence of empty sets. By taking larger $K>0$ and running the same argument, we have $\lim_{s\to\infty}\langle e_s, \phi_{0,s}(f_0)\rangle = \infty$ for every choice of $e, f \in EG$, as desired. $\square$

We have very similar witness lemma for the folding sequence.

Lemma 4.5 (Witness Lemma for Folding Sequence)**.** *Let $(G_s)_{s\geq 0}$ be a reduced folding sequence with every graph isomorphic to G. For $e \in EG$ with $e \in H^i$ for some $i > 0$, there exists an immersed loop α in G such that:*

(1) α contains e,
(2) α does not go through $e' \in EG$ with $e' > e$, and
(3) α does not go through $e' \in EG$ with $e' \sim e$ and $e' \in H^j$ for some $j \neq i$ such that $j > 0$.

Proof. This is the same as Lemma 4.3 except we use $\widetilde{M} := \widetilde{M}^{r,s}$ the normalized transition matrix for the linear map $V_s \to V_r$ induced by the map $G_r \to G_s$. $\square$

4.3. **Upper bound $2n-1$.** Suppose there are k linearly independent ergodic probability measures on the limiting tree of a reduced folding sequence, $(G_s)_{s\geq 0}$, or on the legal lamination of a reduced unfolding sequence, $(G_s)_{s\leq 0}$. Then applying Theorem 1.16 or Theorem 1.17, respectively, we obtain the transverse decomposition $G = H^0 \sqcup H^1 \sqcup \ldots \sqcup H^k$. Now we bound k by carefully measuring the *complexity* of a graph in the following way.

Definition 4.6. Let G be a (possibly disconnected) non-contractible graph and $N_1, \ldots, N_k$ be the *non-contractible* connected components of G. Define the **complexity**, $\chi_-(G)$, of G as:

$$\chi_-(G) := \sum_{i=1}^{k} -\chi(N_i) = \sum_{i=1}^{k} (\mathrm{rk}(N_i) - 1),$$

where $\chi(N)$ is the Euler characteristic of N, which is equal to $1 - \mathrm{rk}(N)$ for a non-contractible graph N.

Now we prove Theorem 4.1 for both folding and unfolding sequences at once.

Proof of Theorem 4.1. Let $A_1, \ldots, A_{k'}$ be the equivalence classes of edges in $G \setminus H^0$ such that if $e \in A_i$ and $e' \in A_j$ with $i < j$, then $e < e'$. Subdivide each A_j further, by letting $B_j^i := A_j \cap H^i$ for each $i = 1, \ldots, k$. It could be that some of B_j^i's are empty. Then take the initial collection I from G as

- All edges in H^0.
- All single-edged loops. Say there are s such loops.
- All rank 1 components of B_j^i's that are not attached to any of H^0, the s single-edged loops, or $B_{j'}^{i'}$ for either 1) $j' < j$ or 2) $j' = j$ and $i' < i$. Say there are m such components.

Then we attach B_j^i's (those that are not part of I) to I one by one in this order:

$$B_1^1, B_1^2, \ldots, B_1^k, B_2^1, B_2^2, \ldots, B_2^k, \ldots, B_{k'}^1, B_{k'}^2 \ldots, B_{k'}^k.$$

We observe that by our choice of the initial collection, and Lemmas 4.3 and 4.5, each addition of B_j^i contributes to the increase of complexity of χ_- at least by 1. Since $\chi_-(G) = n-1$ and $\chi_-(I) \geq 0$, it follows that there can be at most $n-1$ addition of B_j^i's, which means that at most $n-1$ new ergodic measures can be added, as each $B_j^i = A_j \cap H^i$ supports at most one ergodic measure (as it could be empty).

On the other hand, the initial collection I of edges can support at most $s+m$ ergodic measures. This is because H^0 supports none, each single loop can admit at most one ergodic measure, and each B_j^i can support at most one ergodic measure. However, as I has total rank at least $s+m$, it follows that $s+m \leq n$.

All in all, the number k of ergodic measures supported on G is:

$$k \leq (s+m) + (n-1) \leq 2n-1,$$

which proves the upper bound of $2n-1$. The equality is possible only if $s+m=n$ and if each addition of B_j^i exactly increases the χ_- term by 1. □

5. Arationality and Nongeometricity

5.1. **Folding sequence.** Recall that an $\mathbb{R}$-tree T is **arational** if for every proper free factor $H \leq \mathbb{F}$, the action $\mathbb{F} \curvearrowright T$ restricted to H is free and discrete. In this section, we prove that our construction in Section 2.4 can be improved to have the limiting tree be both non-geometric and arational.

Theorem 5.1. *There exists a folding sequence whose limiting tree is non-geometric, arational and admits $2n-6$ linearly independent ergodic length measures.*

We argue that the limiting tree of our folding sequence can be arranged to be arational and free, after perhaps slightly modifying the folding sequence, but without affecting the winning strategy for Alice in the Alice-Bob game, as described in Theorem 2.3.

We endow the folding sequence $(G_i)_{i\geq 0}$ given in Section 2.4 with a metric by assigning arbitrary positive lengths to the edges of G_i's such that they are consistent under pulling back. We then scale these lengths so that the volume of G_0 is 1. Under this metric, all folding maps become isometric immersions on each edge, and such a sequence always converges to an $\mathbb{R}$-tree in $\overline{CV_n}$ by Lemma 1.2.

Take G_t as an induced folding *path* of the folding sequence $(G_i)_{i\geq 0}$, and re-parametrize the folding path G_t so that the integer values of t correspond to the times when illegal turns arrive at a valence ≥ 4 vertex. Thus, t is not an integer if and only if G_t has a vertex with two gates.

We start with the following observation.

Lemma 5.2. *In this pullback metric on the folding sequence $(G_i)_{i\geq 0}$, every edge of G_i has positive length.*

Proof. Suppose that a_1 in some G_i has length zero. Then all edges in G_{i+1} that a_1 maps over also have length 0. From the first matrix in the proof of Theorem 2.5, we see that the edges a_1 maps over include a_1, a_2, b_1, b_2. By the same argument, edges $a_1, \ldots, a_{n-3}, b_1, \ldots, b_{n-3}$ have length 0 in G_{i+n-4}, and then all edges have length 0 in G_{i+n-3}, contradicting that the length function on this graph pulls back to G_0 with volume 1. A similar argument works if any other edge in G_i in place of a_1 has length 0. □

To prove Theorem 5.1, we claim a slightly stronger assertion, but we recall first how we count the number of illegal turns. A gate containing k oriented edges contributes $k-1$ illegal turns. To find the number of illegal turns in a graph, we add those up over all gates.

Lemma 5.3. *Let $H \leq F_n \cong \pi_1(G_0)$ be a proper free factor, or a maximal cyclic subgroup. Then the folding sequence $(G_i)_{i\geq 0}$ can be arranged so that the Stallings core SH_i of the H-cover of G_i has no illegal turns for large i.*

The reason why Lemma 5.3 implies Theorem 5.1 comes from definitions. Indeed, eventually having no illegal turns in the Stallings cores implies that the sequence $(SH_i)_{i\geq 0}$ becomes eventually constant. Since the minimal H-subtree is the limit of the universal covers of the Stallings cores SH_i, it follows that the limiting H-subtree on which H acts has to be a simplicial tree; which is free and discrete. As the same holds for every maximal cyclic subgroup, this implies that there cannot be an elliptic element, showing that the limiting tree is non-geometric.

Proof of Lemma 5.3. Firstly, we observe the number of illegal turns may change only at discrete times. We further claim that in SH_t the number of illegal turns never increases as $t \to \infty$. This is because SH_t immerses into G_t, so a neighborhood of a vertex in SH_t is embedded in that of the image vertex in G_t. If furthermore the immersion $SH_t \to G_t$ is a homeomorphism of the neighborhoods, then the number of illegal turns does not change in this neighborhood at time t. Otherwise, if the map is just an embedding of a neighborhood to a proper subset of the neighborhood of the image vertex, then the number of illegal turns may decrease at time t. Therefore, it suffices to argue that it can be arranged that the number of illegal turns in SH_t *eventually drops*. We remark here that the only ways to increase the number of illegal turns in the folding path are characterized in [3], and a typical scenario is in Figure 5.

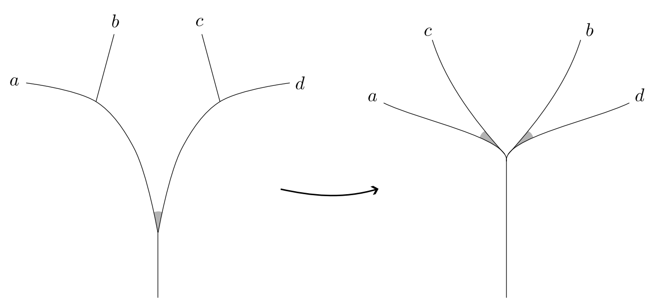

FIGURE 5. (cf. [3, Figure 5].) A typical scenario for the number of illegal turns (colored in gray) to increase after folding. This situation does not happen in our folding sequence; all of our valence ≥ 3 vertices never merge to valence ≥ 5 vertices during the folding.

Now, let v be a vertex in SH_t at which an illegal turn exists. We will insert folding maps in the folding sequence $(G_s)_{s\geq 0}$ such that the number of illegal turns at v in SH_t drops, while ensuring that Alice still has a winning strategy. We divide the cases based on the edges incoming to the vertex v.

Case 1. An illegal turn at v in SH_t enters v through some lift of b_i. We examine what other edges in SH_t are attached at v. If the loop b_i lifts to a *loop* in SH_t at v, then the illegal turn would have entered v on the previous step SH_{t-1}, and we would rewind to that time and move on to the appropriate subcase of Case 2. So, we may assume that b_i does not lift to a loop in SH_t based at v. Then, as H is *root-closed* (i.e., $h^k \in H$ for some $k \neq 0$ implies $h \in H$), no power of b_i lifts to a loop based at v. Let $k \geq 0$ be the maximum k such that there is a path b_i^k in SH_t that starts at v. In SH_t, the edge b_i could form an illegal turn with

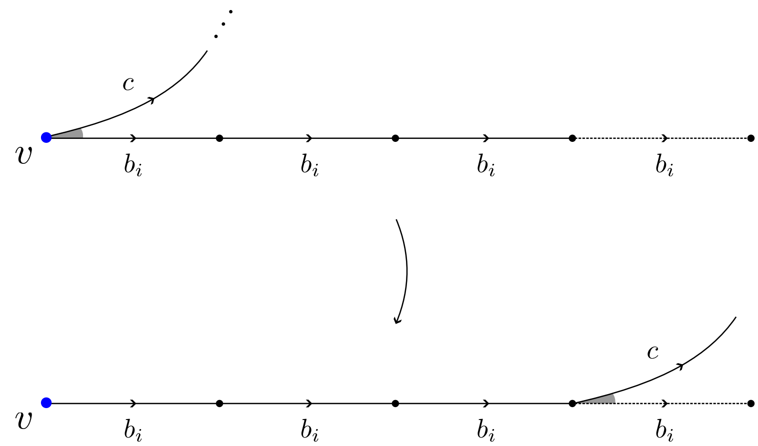

FIGURE 6. Illustration of folding c over b_i by $k = 3$ times as a result of inserting $c \mapsto b_i^3 c \overline{b_i}^4$. There is an illegal turn (in gray) between c and b_i in G_t, but not in the latter *core* since the edge b_i incident to c is not in the core anymore, as drawn in dashed edge.

a_i or $\overline{a_{i-1}}$, as well as with c or $\overline{c}$. If either $\{a_i, b_i\}$ or $\{\overline{a_{i-1}}, b_i\}$ are illegal, we wait; namely, for larger t', we can guarantee that the image of v in $SH_{t'}$ has no illegal turn $\{a_i, b_i\}$ or $\{\overline{a_{i-1}}, b_i\}$. Hence, without loss of generality, it suffices to deal with the case when the turn $\{\overline{c}, b_i\}$ forms an illegal turn at v. Recall that our folding sequence must be constructed in such a way that the resultant transition matrices satisfy the conditions required to preserve Alice's winning strategy, as described in Theorem 2.3. We first try inserting the fold $c \mapsto b_i^k c \overline{b_i}^{k+1}$ in the folding sequence. Then, the number of illegal turns decreases in SH_t, as illustrated in Figure 6.

If k is uniformly bounded throughout the folding sequence, then the proof of Theorem 2.3 shows that Alice still has a winning strategy for this modified folding sequence. Otherwise, we observe that the bounded rank condition on H implies that the number of vertices of valence ≥ 3 in SH_t is bounded. Therefore, on the path b_i^k, we can find a path b_i^s of bounded length $s \geq 0$ from v such that this subpath ends at a valence 2 vertex. In this case, we insert the fold $c \mapsto b_i^s \overline{a_{i-1}} b_{i-1} a_{i-1} \overline{b_i}^s c$. With this fold, the number of illegal turns in SH_t also decreases, as shown in Figure 7. As s is uniformly bounded by $\operatorname{rk} H$, Alice's winning strategy described in the proof of Theorem 2.3 is still valid, and so the resultant folding sequence still satisfies the conditions in List 1. This concludes Case 1.

Case 2. An illegal turn at v in SH_t enters v through some lift of a_i. As in the discussion of Case 1, we can wait if there is an illegal turn $\{a_i, b_i\}$. Hence, it remains to check when the turn $\{\overline{c}, a_i\}$ forms an illegal turn at v. If b_i is not attached as a loop in SH_t at v, then we can convert the illegal turn to $\{\overline{c}, b_i\}$ by inserting $c \mapsto b_i^k c \overline{b_i}^{k+1}$ or $c \mapsto b_i^s \overline{a_{i-1}} b_{i-1} a_{i-1} \overline{b_i}^s c$ as in Case 1. In this case, if b_i is not attached to v at all in SH_t, this conversion results in a decrease of the number of illegal turns in SH_t. Even if b_i is attached to v (but not as a loop), following Case 1, the insertion of such maps will still move the illegal turns to outside the core.

On the other hand, if a_{i-1} is not attached to v in SH_t, then we can insert $c \mapsto \overline{a_{i-1}} b_{i-1} a_{i-1} c$ to first convert the illegal turn to $\{c, \overline{a_{i-1}}\}$. Then, having no a_{i-1} attached to v prevents $\{c, \overline{a_{i-1}}\}$ from being illegal in the core, so we decrease the number of illegal turns in this case as well.

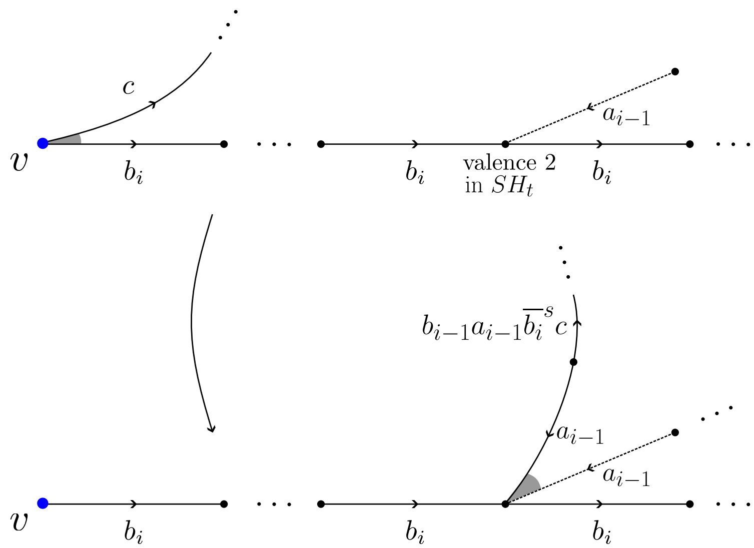

FIGURE 7. Illustration of folding c over b_i by s times as a result of inserting $c \mapsto b_i^s \overline{a_{i-1}} b_{i-1} a_{i-1} \overline{b_i}^s c$. The picture below illustrates that the illegal turn between c and b_i in G_t has already been moved to outside the core, where the dashed edge is an edge not in the core. Hence, the resulting graph after the folding will have less illegal turns in the core.

All in all, we may assume that the core SH_t has both (1) the loop b_i at v and (2) a_{i-1} at v, as well as a_i and c. In this case, we notice that a neighborhood of the vertex v in SH_t is locally homeomorphic to a neighborhood of the image vertex in G_t (after possibly deleting the c loop). Hence, continuing the folding path, c eventually moves along a_i to the next vertex. At this stage, similarly we can either reduce the number of illegal turns, or we see this new vertex is incident with the loop b_{i+1} and a_{i+1}. If the latter case keeps happening, the core SH_t contains a locally homeomorphic copy of $\Delta := \overline{G_t \setminus c}$ (which is isomorphic to Γ' in Figure 3), namely the subgraph of G_t minus the loop c. As the core is compact, it follows that SH_t contains a finite cover of Δ as a subgraph, which implies that H contains a finite-index subgroup of the free factor $\pi_1(\Delta)$. By the root-closed condition on H, it follows that H contains the full factor $\pi_1(\Delta)$. If H is a maximal cyclic subgroup of F_n, then we already have a contradiction by comparing the rank, so consider the case when H is a proper free factor. Here, $\pi_1(\Delta)$ being a maximal proper free factor in $\pi_1(G_t)$ implies that $H = \pi_1(\Delta)$. However, $\pi_1(\Delta)$ keeps changing with folding, as Δ is already legal. This is a contradiction. Therefore, we decrease the number of illegal turns, rather than seeing a copy of G_t minus c inside the core.

Case 3. A 1-gate illegal turn at v in SH_t consists of lifts of c and $\overline{c}$. We note that excluding Case 1 and 2, it remains to consider this case, since every illegal turn either involves c or $\overline{c}$ (or both). For this case, we insert $c \mapsto b_i c \overline{b_i}^2$ in the folding path. In doing so, the number of illegal turns decreases in SH_t so that all turns in SH_t are now legal. Indeed, inserting such a map splits the c and $\overline{c}$ edges so they are now incident to different vertices, as in Figure 8.

To conclude, we note that each time we insert additional folds, the new folding sequence still satisfies the conditions in List 1 required for Alice to have a winning strategy described in Theorem 2.3. Therefore, the resultant folding sequence still

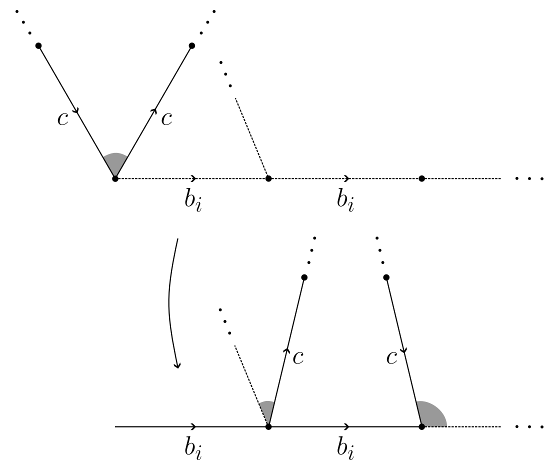

FIGURE 8. Illustration of Case 3. Inserting $c \mapsto b_i c \overline{b_i}^2$ splits the edges $c, \overline{c}$ so that the number of illegal turns in the core decreases.

has transition matrices which satisfy the hypotheses of Theorem 2.4. Furthermore, each time a new fold is inserted for a particular free factor, it does not affect any of the folds or free factors that have been adjusted previously. This is because each time we insert a new fold into the folding sequence, it occurs later in time from any previous fold that has been inserted. Once an illegal turn becomes legal, it stays legal regardless of any additional foldings. □

5.2. **Unfolding sequence.** In this section, we consider the unfolding sequence given in Section 3.4. We show that our construction can be modified to ensure that the legal lamination of our unfolding sequence is arational and non-geometric. In particular, we will consider the inverse of the unfolding sequence given in Section 3.4, and equip it with an appropriate train track structure to make it a folding sequence. Then, the same proof as in Lemma 5.3 shows that we can arrange the unfolding sequence such that its inverse folding sequence has arational limiting tree. This shows that an arational tree is in the accumulation set of the unfolding sequence. Hence, by [5, Theorem 1.1], the unfolding sequence descends into the free factor complex as a quasi-geodesic limiting to a point on the boundary. Therefore, the accumulation points of the unfolding sequence lie in a single simplex consisting of arational trees, concluding that the legal lamination of our unfolding sequence is arational. Thus, it suffices to investigate the inverse of our unfolding sequence and an appropriate train track structure on it.

We first compute the inverses of the rein movers, loopers, and the rotator given in the proof of Theorem 2.5. The inverses of the rein movers $\mathcal{R}_{a_0}, \ldots, \mathcal{R}_{a_{n-4}}$ and $\mathcal{R}_{b_1}, \ldots, \mathcal{R}_{b_{n-3}}$ are:

$$\mathcal{R}_{a_i}^{-1} : \Gamma_{b_{i+1}} \to \Gamma_{a_i}, \qquad \mathcal{R}_{a_i}^{-1} = \begin{cases} c \mapsto \overline{a_i} c a_i, & \\ \mathrm{Id} & \text{otherwise,} \end{cases}$$

$$\mathcal{R}_{b_i}^{-1} : \Gamma_{a_i} \to \Gamma_{b_i}, \qquad \mathcal{R}_{b_i}^{-1} = \begin{cases} c \mapsto \overline{b_i}^2 c b_i, & \\ \mathrm{Id} & \text{otherwise.} \end{cases}$$

Secondly, the inverses of the loopers $\mathcal{L}_{a_1}, \dots, \mathcal{L}_{a_{n-3}}$ and $\mathcal{L}_{b_1}, \dots, \mathcal{L}_{b_{n-3}}$ are:

$$\mathcal{L}_{a_i}^{-1} : \Gamma_{a_i} \to \Gamma_{a_i}, \qquad \mathcal{L}_{a_i}^{-1} = \begin{cases} a_i \mapsto (\bar{b}_i \overline{a_{i-1}}\,\overline{b_{i-1}} a_{i-1})^{\alpha_i} \bar{c} a_i & \\ \mathrm{Id} & \text{otherwise,} \end{cases}$$

$$\mathcal{L}_{b_i}^{-1} : \Gamma_{b_i} \to \Gamma_{b_i}, \qquad \mathcal{L}_{b_i}^{-1} = \begin{cases} b_i \mapsto \overline{a_{i-1}}\,\overline{b_{i-1}}^{\beta_i} a_{i-1} \bar{c} b_i & \\ \mathrm{Id} & \text{otherwise.} \end{cases}$$

Finally, we compute the inverse of the *rotator* ρ:

$$\rho^{-1} : \Gamma_{a_0} \to \Gamma_{a_{n-3}}, \qquad \rho = \begin{cases} a_i \mapsto a_{i-1} & \text{for } i = 0, \dots, n-3, \\ b_i \mapsto b_{i-1} & \text{for } i = 0, \dots, n-3, \\ c \mapsto c. \end{cases}$$

Then our desired map $\mathcal{F}^{-1}$ for the inverse folding sequence is just the inverse of the folding map $\mathcal{F}$ of our unfolding sequence:

$$\begin{aligned} \mathcal{F}^{-1} &= \big(\rho \circ (\mathcal{L}_{a_{n-3}} \circ \mathcal{R}_{b_{n-3}}) \circ (\mathcal{L}_{b_{n-3}} \circ \mathcal{R}_{a_{n-4}}) \circ \cdots \circ (\mathcal{L}_{a_1} \circ \mathcal{R}_{b_1}) \circ (\mathcal{L}_{b_1} \circ \mathcal{R}_{a_0})\big)^{-1} \\ &= \left(\prod_{i=1}^{n-3} [\mathcal{R}_{a_{i-1}}^{-1} \circ \mathcal{L}_{b_i}^{-1} \circ \mathcal{R}_{b_i}^{-1} \circ \mathcal{L}_{a_i}^{-1}] \right) \circ \rho^{-1}, \end{aligned}$$

which is a map from Γ_0 to Γ_0.

To make it a train track map, we endow the train track structure as follows. For oriented edges e_1, e_2 of the graph Γ_e, denote by Γ_{e_1,e_2} the graph Γ_{e_1}(which is defined in Section 2.4), but with different train track structure: it has two illegal turns given by either (1) if neither e_1 nor e_2 is c or $\bar{c}$, then the two illegal turns are given by the two gates $\{e_1, e_2\}$ and $\{c, \bar{c}\}$, or (2) otherwise, the two illegal turns are given by the one gate $\{e_1, e_2\} \cup \{c, \bar{c}\}$ consisting of three oriented edges. See Figure 9 for an example.

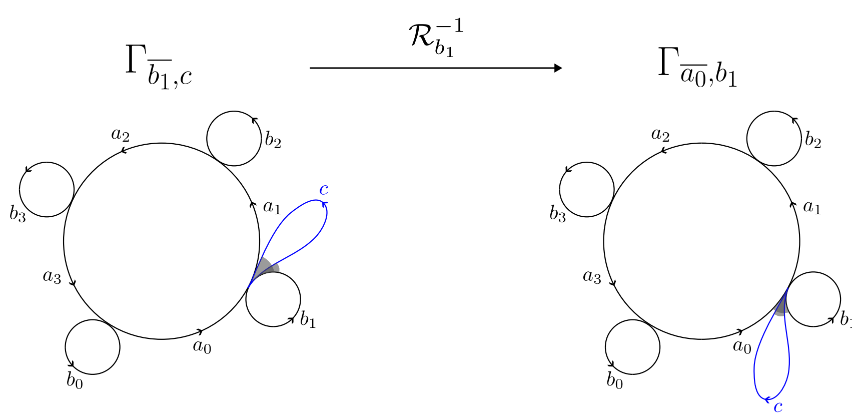

FIGURE 9. An illustration of the inverse $\mathcal{R}_{b_1}^{-1} : \Gamma_{\overline{b_1},c} \to \Gamma_{\overline{a_0},b_1}$ of the rein mover map $\mathcal{R}_{b_1} : \Gamma_{b_1} \to \Gamma_{a_1}$. Here note $\Gamma_{\overline{b_1},c} \cong \Gamma_{a_1}$ and $\Gamma_{\overline{a_0},b_1} \cong \Gamma_{b_1}$ as graphs, but they have different train track structures.

Then we rewrite the domain and codomain of the rein movers and loopers as follows.

$$\begin{aligned}
&\mathcal{R}_{a_i}^{-1} : \Gamma_{\overline{a_i},c} \longrightarrow \Gamma_{\overline{b_i},a_i} && \mathcal{R}_{b_i}^{-1} : \Gamma_{\overline{b_i},c} \longrightarrow \Gamma_{\overline{a_{i-1}},b_i}\\
&\mathcal{L}_{a_i}^{-1} : \Gamma_{\overline{b_i},a_i} \longrightarrow \Gamma_{\overline{b_i},c} && \mathcal{L}_{b_i}^{-1} : \Gamma_{\overline{a_{i-1}},b_i} \longrightarrow \Gamma_{\overline{a_{i-1}},c}\\
&\rho^{-1} : \Gamma_{\overline{b_0},a_0} \longrightarrow \Gamma_{\overline{b_{n-3}},a_{n-3}}.
\end{aligned}$$

Then $\mathcal{F}$ becomes a train track map, making the inverse of our unfolding sequence into a folding sequence. Following the proof of Lemma 5.3, we can show its limiting tree is arational with some insertion of maps to the original unfolding sequence such that the inverses of the inserted maps are folding maps(e.g. inserting $c \mapsto \bar{b}_i^k c b_i^{k+1}$ to the unfolding sequence, has the desired inverse $c \mapsto b_i^k c \bar{b}_i^{k+1}$, which is a folding map with respect to the train track structure on the inverse folding sequence and it is the same insertion as in Case 1 of the proof of Lemma 5.3.) while maintaining the lower bound for the number of ergodic measures. This concludes that the legal lamination of our unfolding sequence is arational as well. Hence we just proved:

Theorem 5.4. *There exists an unfolding sequence whose legal lamination is non-geometric, arational and admits $2n-6$ linearly independent ergodic currents.*

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